HASP Student Payload Application for 2026

Payload Title: Particle Acquisition from Stratospheric Conditions for Analysis in Laboratory (PASCAL)

Institution: University of Chicago

Payload Class (Enter SMALL, or LARGE): LARGE	**Submit Date:** 11/03/2025

Project Abstract:

Particles in the mid-stratosphere remain a large uncertainty in atmospheric models due to the challenges of operating and sampling at high altitudes (>30 km). Characterizing particles in this region is critical, especially with rapid increase in human contaminant levels and the growing possibility of direct material injection into the stratosphere. Now, with the increasing number of smaller and more efficient instruments, particle sampling at these altitudes is now feasible. We propose PASCAL, a modular payload to collect 50 nm to 5 µm diameter, mid-stratospheric particles from the HASP balloon platform for subsequent laboratory analysis using a series of vacuum pumps and cascade impactors. The module will be a low-cost, reduced-weight, and low power consumption payload. These features make the payload appropriate and easily replicated for a range of balloon sizes and research groups. Our goal is to obtain high-quality size-graded data in order to compare analysis techniques (Scanning Electron Microscopy vs Thermal Desorption Mass Spectrometry). Reliable engineering designs will be provided for future instrument development.

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Flight Hazard Certification Checklist

Hazardous Materials List

Classification	Included on Payload	Not Included on Payload
RF transmitters		X
High Voltage		X
Lasers (Class 1, 2, and 3R only) Fully Enclosed		X
Pressure Vessels		X
Non-Rechargeable Batteries		X
Magnets less than 1 Gauss		X
Intentionally Dropped Components		X
Liquid Chemicals		X
Cryogenic Materials		X
Radioactive Material		X
Pyrotechnics		X
UV Light		X
Biological Samples		X
Rechargeable Batteries		X
High-intensity light source		X

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Faculty Advisor Signature: ______________________________

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1 Payload Description

1.1 Payload Scientific / Technical Background

1.1.1 Mission Statement

A long-standing goal of the atmospheric science community is to understand the origin and chemical composition of particles in the middle stratosphere. Yet, they have eluded detailed in-situ characterization and few missions have successfully retrieved samples. As a result, they remain a significant uncertainty in atmospheric models, where they are crucial in stratospheric structure and mixing. This is due to several factors: few balloon-borne missions reaching the required altitude (>30 km), the inability of research aircraft to reach the middle stratosphere, the challenges of operating at pressures as low as 5 mb, and a very low concentration of particles (Testa Jr. et al. 1990).

We propose a modular payload to prototype techniques for collecting 50 nm - 5 μm diameter, mid-stratospheric particles from the HASP balloon platform for a sample return and subsequent laboratory analysis. We aim to build and test a lightweight payload consisting of a Portable Optical Particle Sensor (POPS), various sensors, vacuum scroll pumps and commercial cascade impactors, which will carry both SEM grids and Teflon filters in each impactor stage. Our goal is to validate size-graded sample collection and compare alternate laboratory analysis techniques (Scanning Electron Microscopy and Thermal Desorption Mass Spectrometry) to obtain high-quality scientific data and provide reliable engineering designs for future instrument development. A successful flight and post-flight analysis will: (1) demonstrate the effective use of mini-MOUDI impactors and lightweight pumps at both higher altitudes and lower pressures than previously attempted; (2) characterize the size distribution, morphology, elemental and molecular composition of mid-stratospheric particles collected during the HASP flight. Pre-flight testing and flow experiments will be conducted in collaboration with Dr. Elisabeth Moyer (Department of Geophysical Sciences, University of Chicago). Analysis of the collected particles will be conducted in collaboration with Dr. Phillip Heck, Dr. Mingyi Wang (Department of Geophysical Sciences, University of Chicago), and Dr. Dan Cziczo (Department of Earth, Atmospheric, and Planetary Sciences, Purdue University).

PASCAL will demonstrate a reduced-cost, low-size, weight, and power payload design available for particle collection available for atmospheric science groups. This payload will be designed, assembled, and calibrated entirely by undergraduate students at the University of Chicago, primarily using commercial off-the-shelf (COTS) components modified for stratospheric use. While our mission will only sample at 37km, PASCAL's design can be further applied for future particle characterization missions at and below 37km, therefore contributing to atmospheric characterization well beyond our specific mission.

1.1.2 Mission Background and Justification

1.1.2.1 Mission Science Justification

High-altitude balloons offer the opportunity to study the unique, stable environment of the mid-stratosphere. While particle sampling in the lower stratosphere is dominated by the sulfate aerosols of the Junge layer, the region above 30 km contains a population of particles that is usually inaccessible, containing nano- to micrometer dust of extraterrestrial origins (meteoritic CaO dust and carbon spherules), metallics from spacecraft debris, and rocket exhaust (Testa Jr. et al. 1990), (Lasue et al. 2024). The region also sees some in-situ secondary aerosol formation from gas-phase species, and may contain smoke or soot particles from extreme wildfires or sulfates from volcanoes. These particles are typically grouped into three science categories: Cosmic dust (meteoritic particles), TCAs (terrestrial contaminant artificial particles), and TCNs (terrestrial contaminant natural particles) (Lasue et al. 2024). Collecting mid-stratospheric particles is essential for an unbiased assessment of their relative contributions to current atmospheric models (Testa Jr. et al. 1990).

Cosmic interplanetary dust particles (IDPs) preserve primitive material from the early Solar System, providing insight into Solar System formation processes and the chemical composition of asteroids, comets. Collection of cosmic dust particles at mid-stratospheric altitudes would complement existing lower-altitude datasets (Della Corte et al. 2014). Characterization of anthropogenic particle populations is currently performed only at lower altitudes, and existing databases of TCAs (such as NASA's Cosmic Dust Catalog, Vol 17) have ceased operations. No current missions are routinely sampling at these altitudes. Detailed characterization of stratospheric particles is essential because of rapid changes in the particles in this region. Given the increase in global launch activity, the monitoring of anthropogenic particles is essential to assess the growing impact of human space operations on the upper atmosphere. Similarly, the continued study of TCNs is critical for understanding stratospheric mixing and the transport pathways of terrestrial materials into the stratosphere, including their potential impacts on atmospheric chemistry (Testa Jr. et al. 1990).

High-altitude balloon missions offer the best opportunity to map the nature of the transitory, highly variable population of natural and anthropogenic particles, additionally allowing for characterization of the morphology, size, and chemical composition of each captured particle.

1.1.2.2 Instrument Justification

While a new generation of lightweight instruments have allowed for routine measurements of particle size distributions in the mid-stratosphere (Gao et al. 2016), few campaigns have successfully collected particles for analysis of their chemical composition or morphology. Recent years have seen several attempts at building balloon-borne instruments for particle collection, but all were limited in altitude, collection efficiency, or other characteristics (Table 1). Prior attempts to collect mid-stratospheric particles have been contradictory. Recent measurements from optical particle counters suggest concentrations of ~0.5 particles/cc at 30 km, for radii > 70 nm (Norgren et al. 2024), consistent with sampling in the 1990s that estimated 0.1 particles/cc at 35 km, for radii > 45 nm (Testa Jr. et al., 1990). The DUSTER instrument, flown in 2011, demonstrated the feasibility of non-destructive stratospheric sampling by successfully capturing particles at 30–40 km in the size range of 100 nm to 50 micrometers, but collected only ~1 stratospheric particle per cubic meter.

Mission	Year	Altitude	Particle Size	Sampling Method	Flow rate	Mass	Analytical Instrument
Grawe et al. (HALFBAC)	2021	<15 km	0.1 - 8 μm	Research Aircraft; Vacuum scroll filter sampling	15 L/min	<15 kg	Immersion INP Analysis (LINA, INDA)
Tonson et al. (O-ZONE)	2023	<23 km	0.8 - 2.2 μm	Research Aircraft; Microfiber filter sampling	0.05 L/min	<30 kg	EDX, FTIR, ESEM, GC-MS
Della Corte et al. (DUSTER)	2008 2011 2019 2021	30-40 km	0.2 - 40 μm	HA-Balloon; Cascade Impactor	~15 L/min	<35 kg	FESEM, EDX
J. Testa Jr et al. 1990	1990	34-36km	0.4 - 1 μm	HA-Balloon; Cascade Impactor	N/A	<15 kg	SEM, PIXE, TEM

Table 1. Microparticle Collection Missions and Supporting Details

The goal of this project is to demonstrate that mid-stratospheric particle sampling is possible with a relatively lightweight and simple instrument, allowing for the characterization of a population of particles that is usually inaccessible. The PASCAL instrument will make use of lightweight pumps, electronic components, and compact cascade impactors (Mini MOUDIs) that are now commercially available and widely used in atmospheric science for size-graded particle collection. Mini MOUDIs have been successfully deployed on multiple high-altitude aircraft missions, including SABRE (Stratospheric Aerosol and Gas Balloon Research Experiment) in 2024 and DCOTSS mission (Dynamics and Chemistry of the Summer Stratosphere) in 2022. The HASP campaign will allow us to evaluate the use of mini-MOUDI impactors at the lower pressures and higher altitudes of the middle-stratosphere. Flow tests will be conducted in laboratory settings to evaluate the modified cut sizes for each impactor stage, which should decrease at lower-than-rated pressures, and the optimal flow characteristics. To achieve this, we will employ an environmental chamber along with a generator capable of producing particles ranging from submicrometric to micrometric sizes.

PASCAL will utilize the maximum possible pump capacity for flexibility and to ensure the adequate collection of larger particles. The 6 Scroll Labs SVF-E0-2 pumps can each displace 1.2 L/min at the 5 mb ambient pressure expected on HASP. Using an assumed concentration of 0.2 particles/cc in our size range and operation over the 13-hour float time, the instrument will encounter over one million particles, an order of magnitude more than captured by Testa Jr. in 1990. The HASP campaign will allow us to prototype this sampling strategy.

Beyond our mission-specific sampling goals, the project is intended to demonstrate a reduced-cost, low size, weight, and power (SWaP) design for particle collection that can be used and modified by future research groups. This payload will be designed, assembled, and calibrated entirely by undergraduate students at the University of Chicago, primarily using commercial off-the-shelf (COTS) components modified for stratospheric use. While this mission will only sample at 37 km, the information gathered by the HASP flight can inform sampling strategies at lower altitudes that are accessible by weather balloons. Ultimately, we hope PASCAL's design can form the basis of an efficient lightweight instrument for future general use on weather balloons.

1.1.3 Mission Objectives

Laboratory: testing objectives (Time: pre-integration, before April 30th)

1. Characterize mini-MOUDI pressure drop, cut size, and collection efficiency at lower pressures and higher flow rates than mini-MOUDI rating.
2. Determine flow characteristics for field operation at 5 mb pressure.
3. Evaluate parallel vs. sequential sampling for the 2 sample mini-MOUDIs.
4. Test the reliability of the embedded system in extreme environments (thermal variations, vacuum tests)
5. Test performance of both PASCAL and POPS at 37 km temperature and pressure conditions.

Flight: prototyping engineering objectives (Time: During flight)

1. Demonstrate particles sampling performance at 37 km altitude.
2. Demonstrate size differentiated collection with mini-MOUDIs.
3. Demonstrate the electronic control as well as the safe and reliable operation of PASCAL under harsh environmental conditions.
4. Demonstrate thermal regulation of the PASCAL instrument.

Flight: application objectives (Time: After recovery and lab analysis, December 11th)

1. Assess duration needed for science-quality measurements.
2. Assess collection efficiency by comparison with POPS.
3. Assess ability to collect volatile particles in simple PASCAL setup.
4. Evaluate contamination on descent using control mini-MOUDI.
5. Gather information needed for future iteration to minimize PASCAL weight.
6. Compare and cross-validate different analysis methods.

Science goals (Time: After lab analysis, December 11th)

1. Measure particle size distribution profiles to 37 km using POPS.
2. Measure particle size distribution variation at 37 km using POPS.
3. Collect camera images every 5 minutes.
4. Successfully collect particles from the mid-stratosphere and analyze their morphology and elemental and molecular composition.
5. Determine size-dependence of each class of particles (organics, metallics, soot, cosmic dust).

1.2 Payload Systems and Principle of Operation

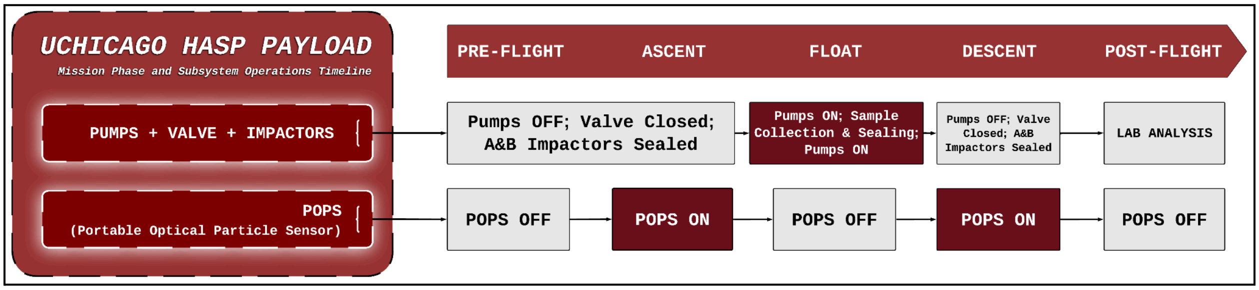


Figure 1. Mission Phase and Subsystem Operations Timeline for the UChicago HASP Payload

PASCAL will consist of three total cascade impactors, two of which will serve as stratospheric particle collectors at float altitude, while the third impactor has no air flow and will act as a control sample. The control impactor will provide a baseline of contaminants collected from the manufacturing/assembly process, mission ascent/descent, and mission operation for subtraction in later analysis. This control impactor is needed because we will not pressure seal our impactors. This setup has been used in other high altitude missions and is implemented because some particle collection missions had significant issues with contaminants unknowingly released by other payload instruments during mission operations (Toson et al. 2023; Testa Jr. et al. 1990). All three cascade impactors are mounted to a payload face that hosts a filter mechanism over the impactors' valves to prevent as many contaminants as possible during transport, deployment, ascent, and descent.

When the HASP flight begins, the first stage of instruments will power on (POPS, camera, sensors, computer). This allows us to test our systems and ensure the flight is operating under normal conditions relative to previous flights to the lower stratosphere. Using the onboard GPS data pulled from the HASP payload telemetry, we will determine once the payload has reached its float altitude of 37 km. At float altitude, an internal mechanism will remove the filters covering the impactors, the vacuum pump system will turn on and begin drawing in particles to the mini-MOUDI impactors, and POPS will shut off. The mini-MOUDI impactors will split the particles based on size onto each stage of the impactor with all small particles being collected in the final filter. The sensors, computer, and camera will operate in parallel with the pumps, providing important contextual information of our sample and the environmental conditions that will be sent in live telemetry as well as stored on the computer for later analysis. The addition of POPS will allow us to compare the lower altitude particle distribution of our flight with previous balloon flights as well as push POPS to higher altitudes than it has previously flown.

Due to the impactors requiring a relatively high flow rate to properly impact particles, we will direct all scroll pumps to one impactor for 6 hours and then use a valve to switch to the other impactor for the remainder of the float period. As the total float time is estimated to be ~13 hours from the HASP Call for Payloads, this will roughly split the particle sample between the two impactors, which allows for comparison between the first and second half of our mission, adding robustness to our measurements. Once HASP GPS data indicates descent has begun, the pumps will be turned off and the filters covering the impactors will be replaced. We will implement redundancy for turning POPS and the vacuum pumps on and off with manual commands from the ground, should downlink telemetry indicate any failures for controlling these systems.

*Figure 2. Image of SABRE modification of Mini MOUDI 135-8B Impactor Plate, showing the inclusion of two filter substrates, a Silicon Nitride Window and Electron Microscope Grid (*Abou-Rizk, 2024)*.*

Following the return of the particle samples, we will conduct laboratory analysis. Similar to Fig. 2, the Mini MOUDIs on PASCAL will include two distinct filter substrates, PASCAL will have different filters than the ones used by the SABRE team. Utilizing instead, SEM grid and Teflon filters in each Mini MOUDI impactor stage for the purpose of distinct analysis techniques. Particles collected on SEM grid filters will be analyzed with a Scanning Electron Microscope (SEM), followed by secondary analysis with Energy Dispersive X-Ray Spectroscopy (EDX). The combination of these two analysis techniques will provide comprehensive information about the collected particles' size distribution, elemental composition, and molecular composition, as well as individual particle chemical composition and morphology. Samples obtained from the Teflon filter, will be analyzed with a Thermal Desorption Mass Spectrometer (TD-MS), a destructive technique that serves to release volatile compounds, which are then analyzed by a mass spectrometer to determine their composition and quantity. TD-MS works to identify the specific organic molecules present in or emitted from collected samples, even having the capability of quantifying specific volatile organic compounds, complementing the elemental analysis provided by the SEM/EDX analysis methods.

1.3 Major System Components

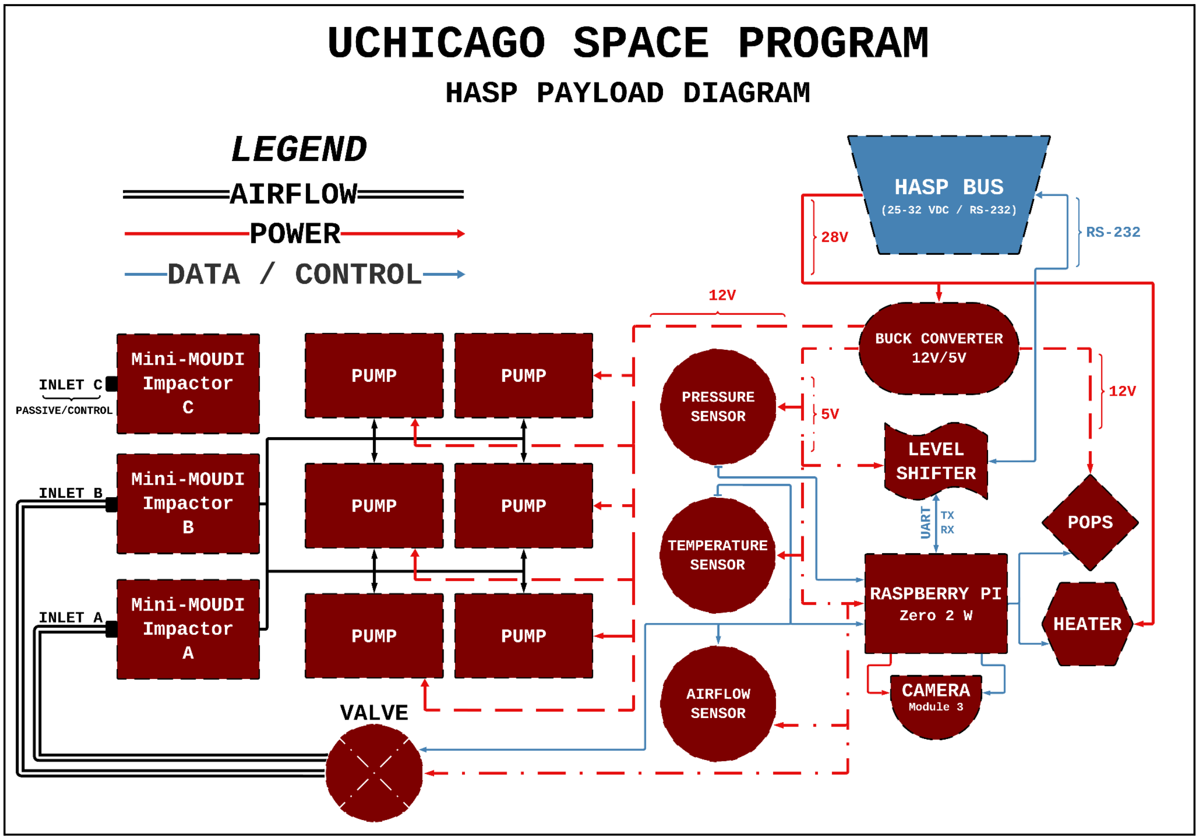


Figure 3. System-Level Block Diagram of the UChicago HASP Payload PASCAL Instrument

The major components of PASCAL are outlined in subsections 1.3.1 through 1.3.5. A detailed description of electronic design and protocols is laid out in Section 1.5.

1.3.1 HASP Interface and Power Conversion

The payload receives 25-32 VDC and an RS-232 link from the HASP bus. A buck converter steps down the bus voltage to three rails: 12 V for the heater and POPS systems, and 5 V for all low-power electronics (Raspberry Pi, sensors, level shifter), as well as the pump array. This block is the sole power source for the payload.

1.3.2 Command, Data, and Telemetry

A Raspberry Pi Zero 2 W is the central computer/controller of the sampling experiment. It polls the pressure and temperature sensors, controls pump and valve activation, and logs POPS data for downlink via the HASP RS-232 link. A level shifter sits between the Pi UART and the HASP RS-232 to translate 3V3 logic to 12 V signals. The Pi is responsible for collecting the status information of operational parameters and sending them to HASP (for live telemetry of pressure, temperature, and POPS sensor readings), and can receive uplink commands. A Pi Camera Module 3 is connected directly to the Pi for visual documentation of the ascent, descent, and surrounding environment. Periodically, a picture will be sent down the downlink in place of telemetry for further atmospheric context from ground.

1.3.3 Particle Sampling Group

The science load consists of three Mini-MOUDI impactors. Impactors A and B are the active samplers. Each active sampler is fed by six ScrollLabs SV-E0-2 pumps (rated for operation at 2 L/min), whose inlets are tied together via a single Pi-controlled valve. The pumps pull ambient air through both impactors selectively and into the pump manifold, providing particulate deposits on the MOUDI stages. The flow rate of the air is measured while pumping and recorded in our telemetry to evaluate the total density of particulates we will have received. To maximize flow rate into the impactors, the valve selectively allows flow into either impactor A or B based on command. Impactor C is a passive control unit: it has no pump connection but will be included at all stages of the build and flight process to isolate possible contamination. This control provides necessary contextual information for proper analysis across all collected samples.

1.3.4 Environmental Sensing

A pressure sensor (5 V) measures ambient pressure for altitude telemetry, critical for valve timing. Eight RTD temperature sensors (5 V) and one built-in temperature sensor report internal thermal conditions on each mini-MOUDI and vacuum pump for both science interpretation and heater control. The Pi reads both and the measured values are included in downlinked telemetry/status packets. The Portable Optical Particle Spectrometer (POPS) provides information on particle size and distribution during ascent and descent, providing important scientific context for our measurements. An airflow sensor (5 V) is used to monitor the flow of the vacuum pumps for our scientific analysis, and may be used for manual control in the event that the automation fails.

1.3.5 Thermal Management

Four 28 V heaters are powered from the converted 12 V rail and are switched/commanded by the Pi. These heaters keep pumps and electronics within operating temperature at float and on ascent/descent. Together, these groups provide regulated power from HASP, centralized control and telemetry, two active particle sampling channels, one passive control sample, environmental context, and thermal survivability, all within HASP electrical and data interfaces.

1.4 Mechanical and Structural Design

Our payload will have dimensions of 32 cm x 28 cm x 30 cm and a weight of ~14.5 kg. The platform contains six pumps, three cascade impactors (with associated flow control, tubing, and solenoid valves), flight electronics, and a POPS instrument (Fig. 25). The general frame structure is made of 1” x 1” aluminum extrusion that is joined with 90-degree gussets using button-head M5 screws and tee nuts. The frame is fastened to the HASP mounting plate with M6 screws tapped into the ends of the aluminum extrusion. Additional mounting screws affixing the bottom horizontal extrusion bars may be added if determined necessary in testing. Aluminum 6061 mounting plates for the pumps and impactors are included for both internal structural strength and ease of assembly during testing. These will interface to aluminum brackets with M5 screws. A 3D printed housing structure, manufactured from thermoplastic polyurethane (TPU) filament, will also be faced onto the impactor mounting plate to minimize stress from the horizontal mount. This will use M5 screws, but use smaller profile hex bolts sunken into a slot in order to prevent interference with the filter mounting bolts. The impactor plate will also be bolted to the interface plate via M6 screws and flange nuts.

The expected greater vertical stresses on ascent and descent necessitated the addition of a horizontal bar of aluminum extrusion affixing the pump plate to the frame via a 90-degree bracket. This extrusion will use gussets as described above, but will also be connected to the vertical extrusions via the pump plate bracket.

All M5 and M6 fasteners used in the payload structure shall conform to ISO 898-1 property class 8.8 or higher. Tightening torques shall be applied in accordance with ISO 16047 or VDI 2230 (Part 1) to achieve 70% of the nominal proof load, unless otherwise specified. Torque shall be applied using a calibrated torque driver in accordance with NASA-STD-5020 Section 4.4. Parts will either be sourced from local suppliers or will be manufactured in-house, at the University of Chicago, by student team members using already available equipment in the student machine shop.

1.5 Electrical Design

PASCAL is monitored and controlled by a single avionics card that manages all Command & Data Handling (C&DH), Power, and Thermal Control functionality. A block diagram of the card is shown in Figure 4 below. **Selected schematics will be included in the following subsections, and a full detailed schematic can be found in Appendix A.1.**

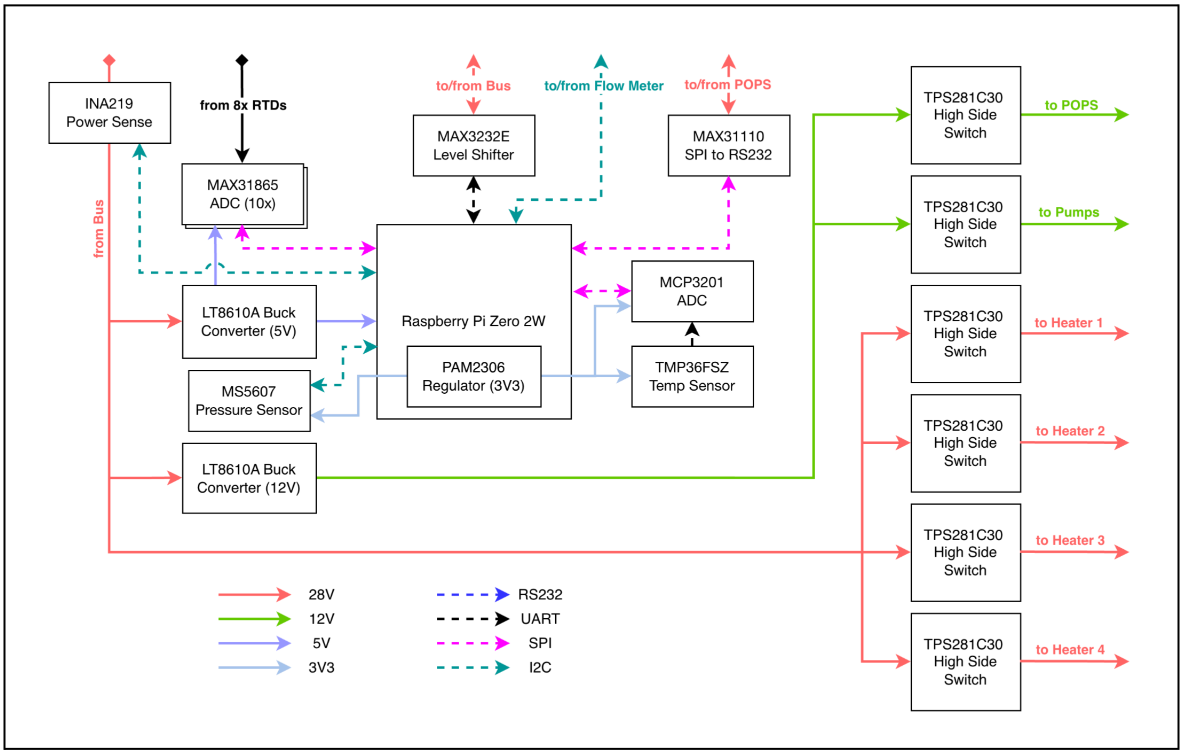


Figure 4: Avionics Card Block Diagram

1.5.1 Bus Interface

The avionics card interfaces with the HASP bus through two female wire-to-board connectors.

The first connector is a 2-pin Molex Pico-Lock, wired to the leads of the EDAC-516-020 receptacle on the mounting panel, pictured in Figure 5. The connector is exclusively used for power delivery, with the analog and discrete pins remaining unconnected. Pin A supplies 26-30 V (nominally 28 V) at 2.5 A maximum to the avionics card for distribution (see Power section below) and is connected to pin 1 of the Pico-Lock. Pin X is the ground pin for PASCAL, connected to pin 2 of the Pico-Lock. The Pico-Lock connector is rated at greater than 2.5 A.

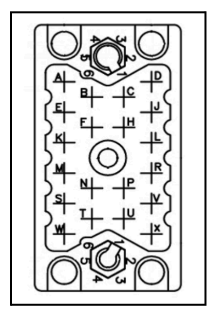
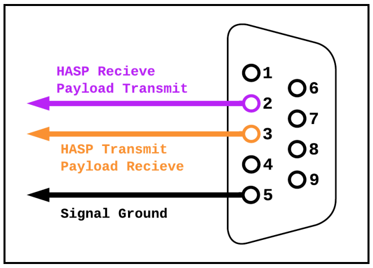


Figure 5: EDAC-516-020 pinout, sourced from the HASP Interface Manual



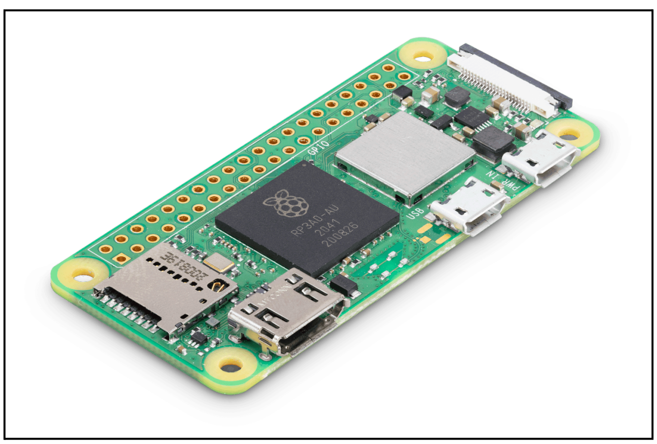
The second connector is a 4-pin Molex Pico-Lock, wired to the leads of the 9-pin D-Sub connector on the mounting panel, pictured in Figure 6. This connector is responsible for the RS-232 serial communication between the avionics card and the bus. The electrical connections are described in Table 2 below.

Figure 6: DB9 pinout from the HASP Interface Manual

Pico-Lock Pin Number	D-Sub Pin Number	Description
1	2	RS-232 Payload TX, Bus RX
2	3	RS-232 Payload RX, Bus TX
3	NC	NC
4	5	GND

Table 2: Pico-Lock to DB9 pinout

1.5.2 Command & Data Handling



Command and Data Handling (C&DH) is performed by a single Raspberry Pi Zero 2 W single-board computer (SBC), mounted directly on the avionics card to minimize system complexity. The Raspberry Pi was chosen for its straightforward development environment (allowing all programming to be done in Python) and its native compatibility with the Raspberry Pi Camera Module 3. The Zero 2 W also has an integrated MicroSD slot for data storage of POPS, all sensors, and captured images.

Figure 7: Raspberry Pi Zero 2 W

The Zero 2 W interfaces over RS-232 to the bus via a MAX3232E level shifter (circuit schematic in Figure 8), enabling the Zero 2 W to use its internal UART to natively communicate with the serial system.

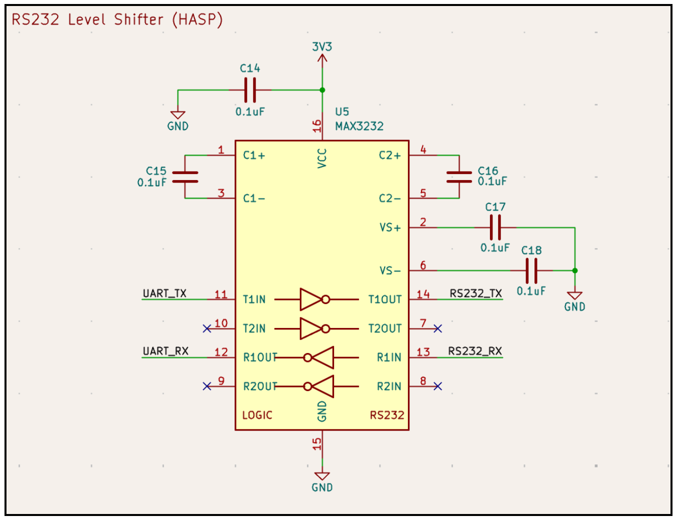


Figure 8: UART to RS-232 Level Shifter Schematic

1.5.3 Power

Components within PASCAL operate on three different voltage rails: 12V, 5V, and 3.3V (henceforth 3V3).

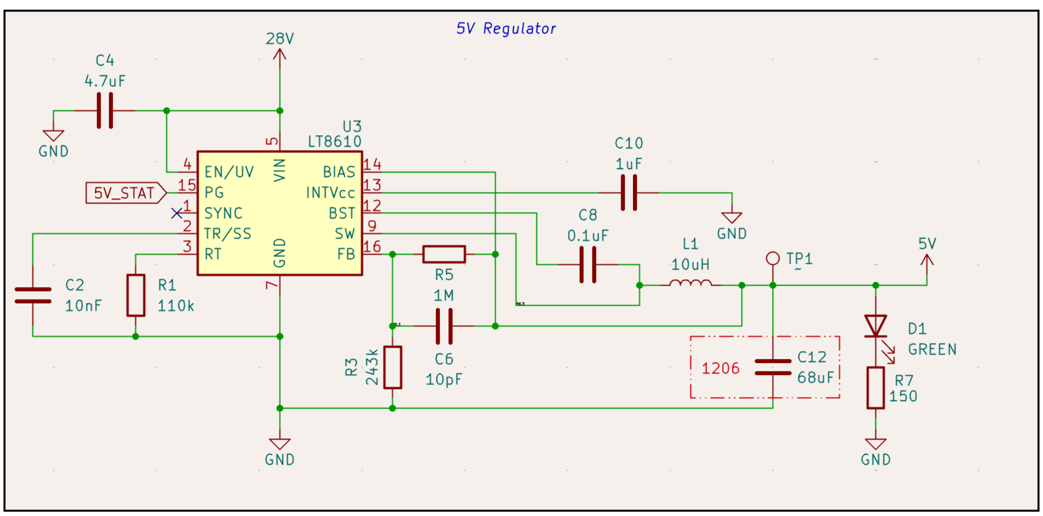


Figure 9: 5V LT8610A buck converter schematic

The 12 V and 5 V voltage rails use LT8610A buck converters at 400 kHz switching frequency due to their high efficiency, low output voltage ripple, and low EMI. The regulators are capable of delivering up to 3.5 A per rail with greater than 90% efficiency. The 5 V buck converter schematic can be found in Figure 9, with all schematics found in Appendix A.1. The UChicago Space Program has experience with these regulators, which are flying on the team's flagship CubeSat, PULSE-A. Existing hardware in the lab is shown in Figure 10.

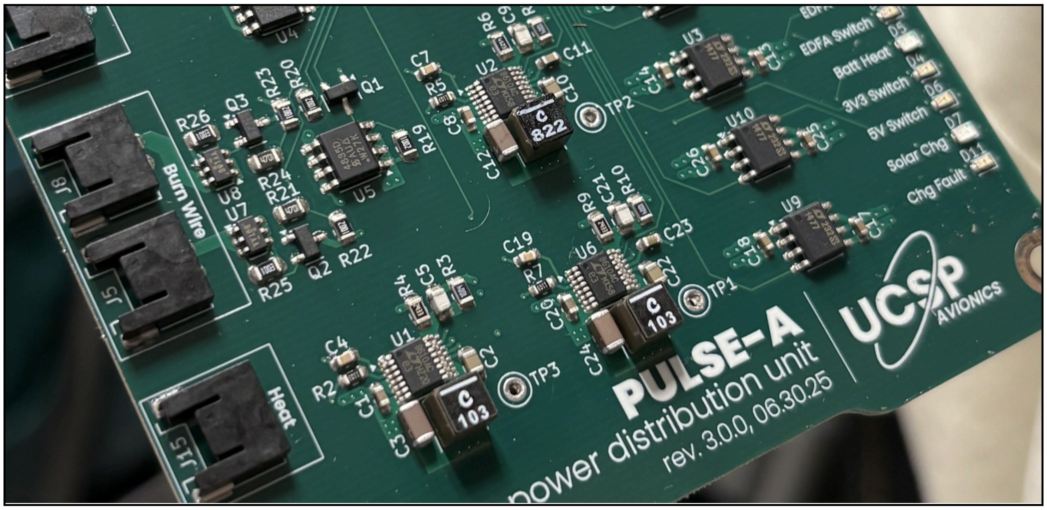


Figure 10: Existing UChicago Space Program hardware with LT8610A voltage regulators

The 3V3 line is supplied by the Raspberry Pi Zero 2 W's internal 3V3 buck regulator (part number PAM2306AYPKE) which is capable of delivering 1A continuous output. The 3V3 line supplies the onboard ADCs and sensor ICs, so its low current capacity is sufficient.

The 12 V line supplies the vacuum pumps as well as the POPS instrument. As previously mentioned, the POPS instrument is only operated during ascent/descent, and the vacuum pumps are only operated during float. The pumps are driven by one TPS281C30 single-channel high-side switch, which can be controlled by the Zero 2 W (Figure 11). The TPS281C30 has an adjustable current limit and integrated thermal shutdown capability to prevent overdraw by the vacuum pumps in an anomalous situation, preventing the payload fuse from blowing to maintain payload operation. The POPS instrument is powered by the 12 V rail through a second TPS281C30 switch. The switches will be operated such that neither will be on at the same time, as dual operation would require too much power.

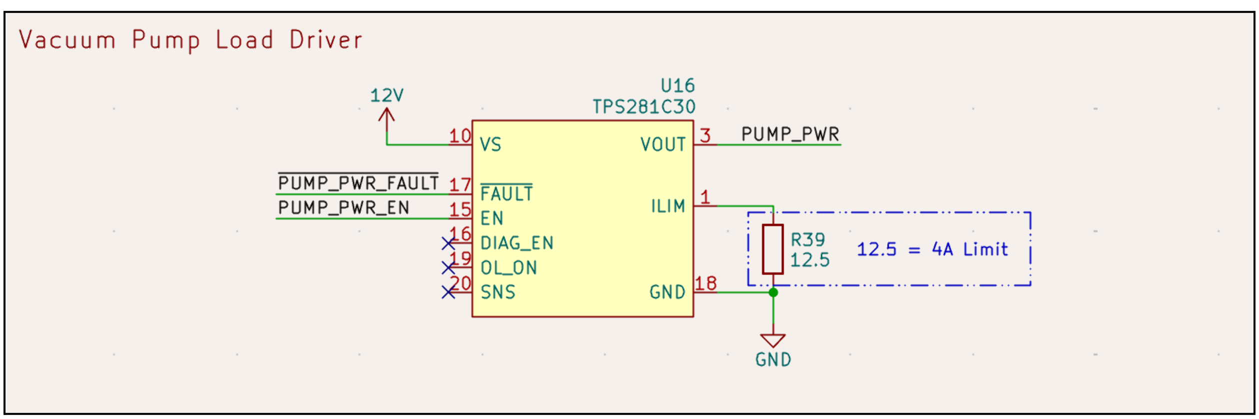


Figure 11: Vacuum Pump Load Driver Circuit

PASCAL's power draw remains below the maximum supplied by the bus with 1.5 W margin, accounting for intrinsic distribution inefficiencies with power regulation. A detailed breakdown can be found in section 3.2.

The heaters are driven by identical load drivers to the vacuum pumps except utilizing the 28 V line for reduced current draw, each with its own driver to allow for distributed heat control based on RTD feedback.

Additionally, there is an INA219 current sensor with a shunt resistor placed immediately after the connector to the HASP bus (Figure 12), allowing for precise measurement of power draw by PASCAL. This prepares for an emergency shutdown in the event of a potential fuse blow.

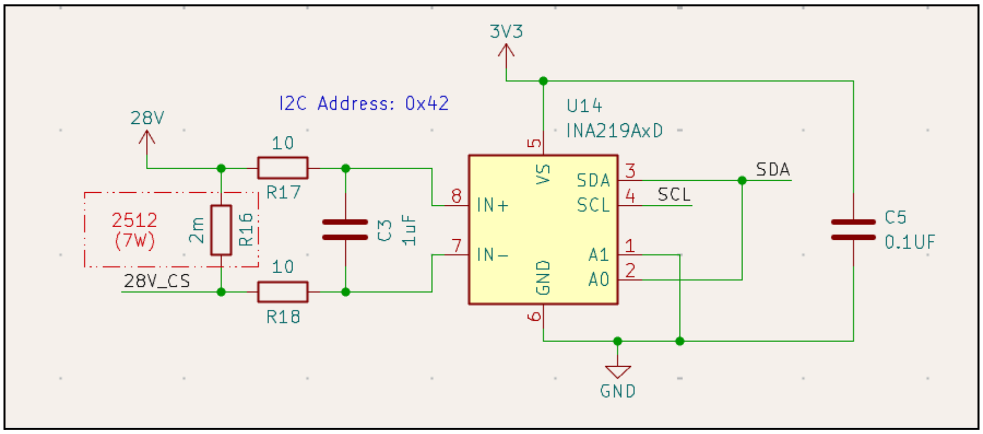


Figure 12: Current Sense and shunt resistor setup. "28V_CS" is the line direct from the DB9 connector, with 28V then distributed to the system.

1.5.4 Sensor Design

PASCAL's sensors are sorted into two categories: those mounted to the avionics card and those that are distributed throughout the payload.

The following sensors are mounted directly to the avionics card:

Pressure Sensor (MS560702BA03-50):
The MS5607 is a 24-bit-precision board-mounted absolute pressure sensor, qualified for 1kPa to 120kPa. This will be used for context measurements of the flight during ascent and descent. The sensor interfaces over I^{2}C with the Zero 2 W.

Figure 13: MS5607 Pressure Sensor IC

PCB Temperature Sensor (TMP36FSZ):

The TMP36FSZ analog temperature sensor will measure the temperature of the avionics card itself and is accurate to within ±2°C. An MCP3201 12-bit Analog-to-Digital Converter (ADC) will be used before the Zero 2 W reads the data due to the lack of an internal ADC on the SBC.

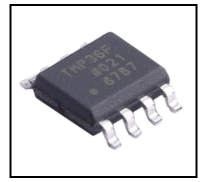


Figure 14: TMP36FSZ Pressure Sensor IC

The following sensors are external to the card:

Ambient Temperature Sensor (PT100):

The PT100 is a high-precision resistance temperature detector (RTD) to detect the ambient temperature of the balloon. NOAA has flown PT100 arrays and has unparalleled consistency. These sensors will be used in closed feedback loops with the heaters to regulate the balloon's temperature. Each sensor's analog output is converted to digital with a MAX31865 RTD-to-digital IC. Then, the data is sent over SPI to the Zero 2 W.

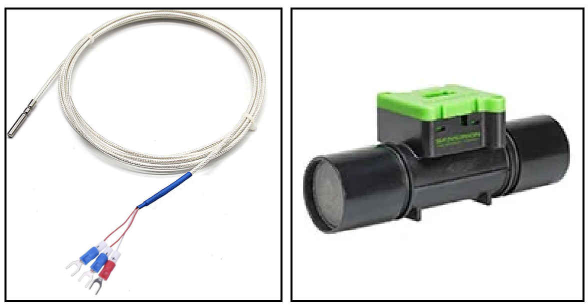


Figure 15: PT100 Temperature Sensor and SFM3000-200C Flow Meter (Left to Right)

Pump Flow Sensor (SFM3000-200C):

The SFM3000-200C is a high-precision gas flow sensor accessible to the computer via I^2C. It fits between the vacuum pumps and the filter and will allow us to monitor and control the flow of the pumps.

Portable Optical Particle Spectrometer (POPS):

POPS data will be logged by the Raspberry Pi Zero 2 W via RS-232 serial communication. A MAX3110 SPI/RS-232 bridge will enable the Zero 2 W to communicate over RS-232 with POPS. The instrument was initially designed by NOAA, who has offered to loan the instrument to us. The measurements will be performed and processed independently on the computer provided in POPS, then reported to our avionics card to provide external measurement of particle size distributions. At lower altitudes, we can compare the POPS data we receive to data from missions the system has previously flown on. It has been used for altitudes up to 28 km. Our mission will also test the sensor up to less than 10 hPa and 35 km, stretching the well-known system to its limits to provide critical information to when the spectroscopy system begins to degrade.

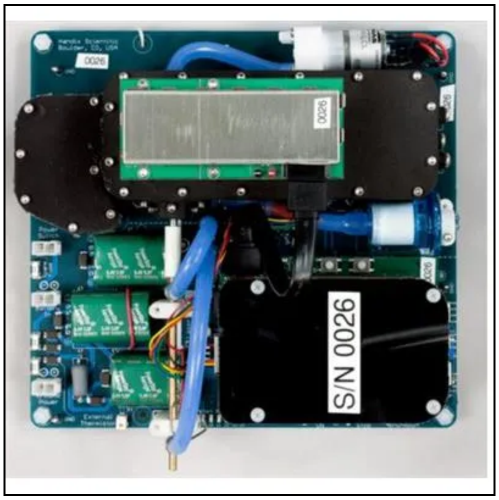


Figure 16: Overhead view of Portable Optical Particle Spectrometer (POPS) (Mei et al. 2020)

1.6 Thermal Control Plan

1.6.1 Thermal Environment

The thermal environment of the middle-upper stratosphere presents extreme conditions that significantly impact the payload system design. At altitudes between 30-50 km, ambient air temperatures range from approximately -3°C and -15°C, an increase from the tropopause's minimum of approximately -56°C due to ozone layer heating (Standard Atmosphere, 1976). Convective heat transfer at these altitudes is nearly negligible; instead, the dominant thermal process experienced by high-altitude balloons is radiative heating from unfiltered solar radiation. Radiative heating creates a harsh environment in which sun-facing surfaces can reach 0°C, while shadowed surfaces can drop to -70°C, resulting in large thermal gradients across payload systems (NOAA, Radiation Primer).

1.6.2 Component Considerations

Component	Operating Range
Mini-MOUDI	10°C to 50°C
Vacuum Pump	-40°C to 50°C
Raspberry Pi Zero 2W	-20°C to 70°C
Board-mounted pressure sensor	-40°C to 85°C
Air flow sensor	-20°C to 80°C
Camera	0°C to 50°C
POPS	0°C to 70°C

Table 3: Payload component operating temperatures

Note: The diode laser and the PMT in the POPS are not temperature-regulated, leading to temperature-dependent variations in laser intensity. Thus, to achieve the best performance, we will minimize thermal variations using heaters and radiative-fin heatsinks.

1.6.3 Thermal Control

Thermal control will be handled by both the flight software on board and the sensors described above. The computer will control the heating of four 5 W Kapton Heaters to prevent component freezing. All major heat sources (pumps, electronics) will contain radiative-fin heatsinks for cooling. A closed loop control scheme will be managed on a per-sensor basis by the on computer, enabling precise control of different areas of PASCAL. To safeguard against component overheating at altitude, the system will default to off for all heaters if any component exceeds its rated temperature. Controlled thermal testing will take place in vacuum chambers at precise pressures, enabling us to confirm that heat sources do not interfere with each other: if they do, a more robust, parameterized model will be developed.

2 Team Structure and Management

2.1 Team Organization and Roles

Portrayed in Figure 17 are the tentative roles and responsibilities of each team member. The team will meet weekly, either in person at the University of Chicago or virtually over Zoom, until the HASP launch.

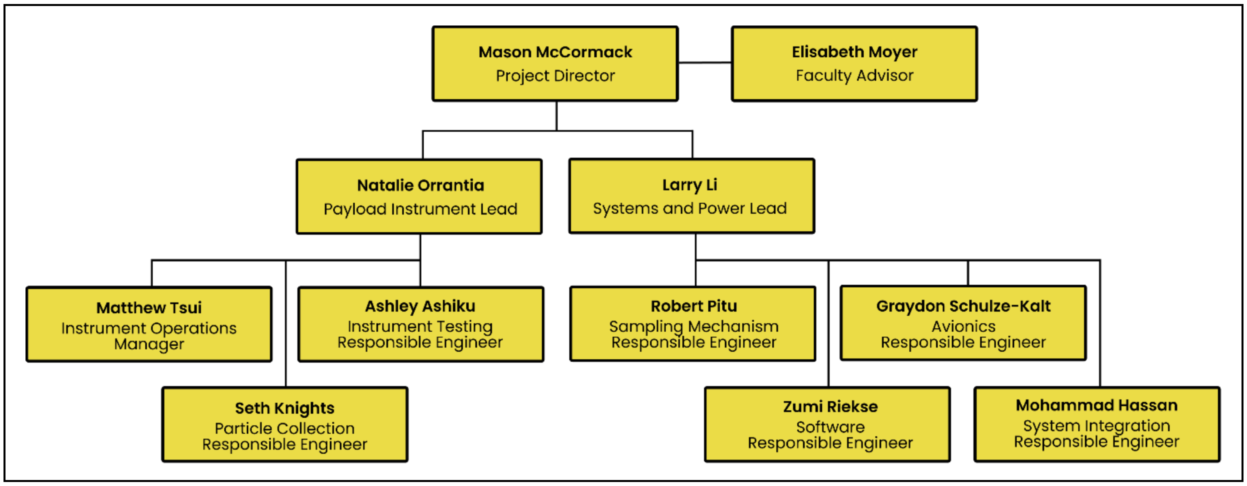


Figure 17: PASCAL Team Organization Chart

Name	Role	Responsibilities	Major	Year
Mason McCormack masonmccormack@uchicago.edu (857) 283-7264	Project Director	Manage development of payload, calibration of instruments, meeting deadlines, and coordination with HASP team	B.S. Astrophysics	U.G. 3rd
Natalie Orrantia orrantianm@uchicago.edu (772) 626-3712	Payload Instrument Lead	Ensuring payload meets science goals, coordinating with labs for sample return analysis.	B.S. Astrophysics B.A. Physics	U.G. 4th
Lawrence Li lkli@uchicago.edu (512) 363-9455	Payload Systems Lead	Manage payload structure and manage CAD to fit payload objectives	B.S. Physics B.S. Computer Science	U.G. 3rd
Mohammad Hassan mohassan@uchicago.edu (708) 714-6537	Systems Integration Responsible Engineer	Manage the integration of various subsystems on payload	B.S Molecular Engineering	U.G. 3rd
Ashley Ashiku aashiku@uchicago.edu (925) 286-2102	Instrument Testing Responsible	Conduct laboratory testing on instruments	B.S. Astrophysics	U.G. 3rd

	Engineer	and modify to meet payload objectives		
Danielle Riekse dzriekse@uchicago.edu (703) 795-1538	Software Responsible Engineer	Develop software for embedded systems on board for operation and testing	B.S. Computer Science B.S. Physics	U.G. 2nd
Matthew Tsui matthewtsui@uchicago.edu (708) 965-3383	Instrument Operations Manager	Procure and manage payload instruments throughout development to testing and flight	B.S. Economics	U.G. 4th
Robert Pitu robertpitu@uchicago.edu (872) 212-0472	Sampling Mechanism Responsible Engineer	Develop, manage, and calibrate payload design to enable maximum sampling efficiency and proper contextual measurements	B.S. Astrophysics B.S. Molecular Engineering	U.G. 3rd
Graydon Schulze-Kalt graydonsk@uchicago.edu (310) 978-7535	Avionics Responsible Engineer	Develop PDU and avionics stack, manage power of all components at both stages.	B.A. Physics	U.G. 4th
Seth Knights knightss@uchicago.edu (971) 707-9423	Particle Collection Responsible Engineer	Manage, design, and calibrate impactor and pump system, including any necessary modifications	B.S. Physics	U.G. 4th

Table 4: Demographic information and responsibilities of team members

2.2 Timeline and Milestones

Below we present a timeline of PASCAL's development, culminating with a projected launch in August 2026.

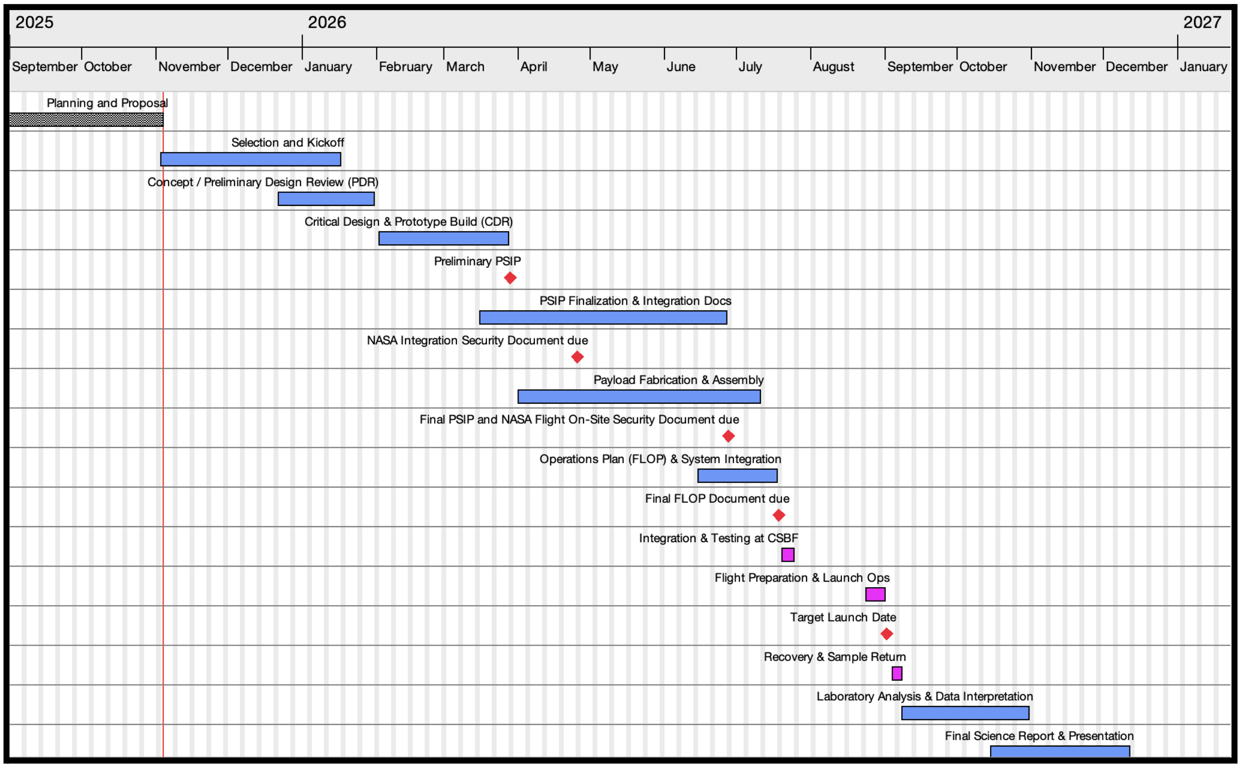


Figure 18: Project Gantt Chart

Detailed Timeline:

Planning and Proposal *August 22, 2025 - November 3, 2025*	Drafted proposal. Defined mission objectives. Conducted literature review, initial CAD concept, initial electrical schematics and budget. Participated in Q&A Telecon **(Sept 26)** and Application Dev Telecon **(Oct 17)**. Submit a full proposal by November 3, 2025.
Selection and Kickoff *November 3, 2025 - January 16, 2026*	Week 1: After submission, finalize concept sketches and begin component sourcing for testing. Weeks 2-3: Review feedback expectations; outline major work packages Week 4: Draft mechanical sketches and power interface Week 5: Following selection **(Dec 5)**, finalize payload configuration, finalize power budget Weeks 6-8: Devise and draft funding proposals Weeks 9-11: Present funding proposals to academic departments and prospective sponsors. Confirm final budget and material acquisition strategy. Attend HASP Kick-Off Meeting **(Jan 16)**.
Concept / Preliminary Design Review (PDR) *December 22, 2025 - January 30, 2026*	Weeks 8-11: Conduct CoDR/PDR for sensors, electronics, and structure. Develop schematic for pump, filters, and data logger. Simulate mass/power budgets. Weeks 12-15: Address PDR feedback; order prototype parts
Critical Design & Prototype Build (CDR) *February 2, 2026 - March 27, 2026*	Weeks 16-19: Fabricate PCB prototype; 3D print enclosure. Begin flow-rate testing. Weeks 20-22: Electrical bench testing; verify telemetry output Week 23: Preliminary PSIP **(Mar 27)**.
PSIP Finalization & Integration Docs *March 16, 2026 - June 26, 2026*	Week 28: Submit NASA Integration Security Doc **(Apr 24)** Weeks 29-33: Iteratively complete PSIP sections while fabrication/testing proceeds. Week 34: Submit Final PSIP and NASA Flight On-Site Security Document **(Jun 26)**.

Payload Fabrication & Assembly *April 1, 2026 - July 10, 2026*	Weeks 24-28: Machine structure; mount mechanical subsystems. Weeks 29-32: Integrate electronics; perform power and continuity checks. Weeks 33-35: Thermal and vibration testing; flight-configuration review. Weeks 36-39: System-level tests; verify telemetry; finalize build. Full build: CNC panels, mount pumps/filters, wire harness. Conduct electrical, vacuum, and thermal tests. Update as-built drawings for FLOP.
Operations Plan (FLOP) & System Integration *June 15, 2026 - July 17, 2026*	Weeks 33-36: Draft FLOP; define operational timeline. Draft and validate flight procedures, telemetry protocols, and recovery plan Weeks 37-38: Dry-run telemetry; collect test data. Weeks 39-40 Submit FLOP **(Jul 17)**; finalize packing list. Conduct internal Flight Readiness Review.
Integration & Testing at CSBF *July 20, 2026 - July 24, 2026*	Week 41: On-site payload integration in Palestine, TX. Electrical and communication verification with HASP.
Flight Preparation & Launch Ops *August 24, 2026 - August 31, 2026*	Weeks 42-45: Inspect post-TVAC hardware; run rehearsals Week 46: Participate in HASP flight prep **(Aug 24-25)** in Fort Sumner New Mexico, perform system checks, and support. Launch **Aug 31**.
Recovery & Sample Return *September 4, 2026 - September 7, 2026*	Week 47: Retrieve payload, inspect filters, and ship to lab. Document post-flight condition
Laboratory Analysis & Data Interpretation *September 8, 2026 - October 30, 2026*	Weeks 48-49 Filter handling; spectroscopy analysis. Weeks 50-51: Correlate altitude, telemetry, and sample results. Weeks 52-54: Statistical analysis; draft figures and conclusions.
Final Science Report & Presentation *October 15, 2026 - December 11, 2026*	Weeks 52-55: Write final report; integrate plots and findings. Weeks 56-57: Internal and faculty review. Week 58: Submit Final Science Report.

Table 5: Detailed timeline of PASCAL team operations

Every First Friday: To attend the HASP Zoom Video Conference.
Every Last Friday: Submit HASP monthly status report.

2.3 Anticipated Participation in Integration and Launch Ops

At least three student team members from the University of Chicago, as well as the faculty advisor, will travel to the Columbia Scientific Balloon Facility (CSBF) in Palestine, Texas, for payload integration and environmental qualification testing during the dates provided by the HASP team.

At least three students and the faculty advisor are also expected to participate in launch operations at Fort Sumner, New Mexico, on the official dates established by HASP and CSBF. All personnel will complete the required safety, handling, and integration training prior to on-site work.

2.3.1 Anticipated Procedures

Prior to Integration

1. Complete final bench and environmental testing of the particle sampling system, including flow-control hardware, sensors, and onboard electronics.
2. Verify current draw, grounding, and connector integrity under simulated HASP 28 V DC power
3. Perform software validation to ensure stable data logging, timestamp accuracy, and telemetry formatting.
4. Inspect and torque all structural fasteners; apply vibration-resistant locking compounds.
5. Prepare and review integration documentation, including updated wiring diagrams, test results, and PSIP appendices.

Integration at CSBF

1. Mount the payload securely to the assigned HASP gondola interface plate using approved hardware.
2. Connect power and serial/data interfaces according to verified configuration drawings.
3. Run the full command sequence to test autonomous sampling logic, sensor initialization, and storage redundancy.
4. Monitor live data for flow stability, temperature regulation, and system health parameters.
5. Record all observations and coordinate any troubleshooting with CSBF engineering personnel.

Thermal-Vacuum Testing and Flight Readiness

1. Participate in the official CSBF thermal-vacuum qualification.
2. Validate sensor stability, data integrity, and thermal performance under simulated float conditions.
3. Confirm mass, balance, and envelope compliance with HASP mechanical constraints.
4. Review telemetry output with HASP operations to verify uplink and downlink reliability.
5. Obtain final certification for flight configuration prior to shipment to Fort Sumner.

Launch Operations (Aug 31, 2026)

1. Support pre-flight continuity, power, and communication checks on the flight line.

2. Coordinate with HASP personnel during ascent and float to verify telemetry reception.
3. Activate and monitor sampling sequence at target altitudes.
4. Record real-time telemetry for pressure, pump current, and temperature to confirm nominal operation.

Post-Flight and Recovery

1. Retrieve and secure collected filter media or captured particle samples for laboratory analysis.
2. Perform post-flight inspection of electronics, connectors, and data storage devices.
3. Transfer all data to archival storage and begin laboratory processing and analysis of collected samples.

3 Payload Interface Specifications

3.1 Weight Budget

Item	Mass (g)	Uncertainty	Comments
mini-MOUDI Impactors (3)	~225*3 = 675	Medium (30%)	Modifications may change mass slightly
Vacuum Pump (6)	~250*6 = 1500	Low (10%)	Little modification, mass standardized
POPS	~800	Low (10%)	
Heater (4)	~90*4 = 360	Low (10%)	
Valve (2)	~1041*2=2082	Low (10%)	Little modification, mass standardized
Computer + Sensors + Camera + Board	~30	Low (10%)	Little modification, mass standardized
Structures, pipes, and wires	9116	Medium (30%)	Subject to change as design improves
Total	**14563**		

Table 6: Weight budget for components of PASCAL

3.2 Power Budget

Note that during float POPS will be turned off and the vacuum pumps will be turned on, and vice versa during ascent. As such we present two power budgets for the two different power stages of the mission to adequately represent a flight-like power budget.

3.2.1 Ascent/Descent

Item	Voltage	Current	Uncertainty	Power (W)	Comments
Computer + sensors	5 V	0.2 A	20%	1.2 W	Sourced from RaspiTV
Camera	5 V	0.15 A	20%	0.90 W	Sourced from RaspiTV
POPS	12 V	0.4–0.5 A	10%	7.7 W	Source linked
Heater (4)	26 V	0.19 A		21 W	Heater voltage is based on the worst-case supply voltage. Expected nominal voltage of 28 V.
Distribution Losses	26 V	0.05 A	20%	1.6 W	Estimated from 94% buck converter efficiency
TOTAL				**32.34 W**	

Table 7: Power budget for ascent and descent portions of mission

3.2.2 Float

Item	Voltage	Current	Uncertainty	Power (W)	Comments
Pumps (6)	12 V	0.5 A	5%	37.80 W	From datasheet
Computer + sensors	5 V	0.2 A	20%	1.2 W	Sourced from RaspiTV
Camera	5 V	0.15 A	20%	0.90 W	Sourced from RaspiTV
Heater (4)	26 V	0.19 A	5%	21 W	Heater voltage is based on the worst-case supply voltage. Expected nominal voltage of 28 V.
Distribution Losses	26 V	0.10 A	20%	3.1 W	94% efficiency.
TOTAL				**63.52 W**	

Table 8: Power budget for float portion of mission, at 37 km altitude

3.3 Downlink Serial Data

By default, the standard telemetry data packet will be downlinked every 10 seconds. If the instrument is prompted to take a picture, it will instead downlink the image data packet, and will not downlink telemetry until the image is successfully sent. Data will be sent through the balloon's onboard transmitter, which is connected to the payload via the DB9 connector at a band rate of 4800, consistent with the transmission rate.

3.3.1 Data Packet (Static 54 bits):

Byte	Title	Description
1-2	Header	Simple '\x1\x21' header indicating the start of a new packet of data
3	ID	ID for telemetry packet
4-12	Timestamp	Simple 8 byte timestamp
13-16	Altitude	GPS altitude data from -9999.9 to 99999.9 (m)
17-24	Temperature	Temperature sensor data (deg C)
25-28	Pressure	Pressure data (hPa)
29-30	Pump air flow rate	Air flow rate from the sensor on the pump when floating
31-32	Pump air flow rate	Redundancy
33-34	Particle density 1	POPS data on particle density
35-36	Particle density 2	Redundancy
37-38	Flow rate	POPS data on flow rate
39-40	Flow rate	Redundancy
41-42	Particle size	POPS data on particle size
43-44	Particle size	Redundancy
45-46	Motherboard temperature	Temperature from the thermistor on the motherboard
47	Voltage Regulator Status	Status of voltage regulators on the computer
47-52	checksum	

53-54	Terminator	Simple '\x3\xD' terminator indicating the end of a packet of data

Table 9: Standard data packet of telemetry

3.3.2 Image Packet (compression on average achieves ~200 KB)

Byte	Title	Description
1-2	Header	Simple '\x1\x21' header indicating the start of a new packet of data.
3	ID	ID for image packet
3-11	Timestamp	8 byte timestamp
12-13	Image Size	Size of image
14-199998	Compressed Image Data	All image data (size variable)
199999-200000	Footer	Simple '\x3\xD' terminator indicating the end of a packet of data

Table 10: Standard image data packet from the Pi Camera Module

3.4 Uplink Serial Commanding

We will use uplink commands that follow the standard format up to the HASP specifications for uplink transmission. The two command bytes are IDs for each command and will be interpreted by the on-board motherboard to fulfill the necessary function.

Command ID	Command Code Byte 1	Command Code Byte 2	Description
RESET	0x01	0x10	Reset computer
TAKE_PHOTO	0x13	0x31	Take a photo
TURN_ON_PUMP	0x15	0x03	Manual backup: turns on the vacuum pump
TURN_OFF_PUMP	0x02	0x18	Manual backup: turns off the vacuum pump
TURN_ON_POPS	0x50	0x01	Manual backup: turns

			on POPS computer and system
TURN_OFF_POPS	0x05	0x93	Manual backup: turns off POPS system

Table 11: Possible uplink commands and corresponding command codes

3.5 Analog Downlink

N/A

3.6 Discrete Commanding

N/A

3.7 Payload Location and Orientation Request

We prefer to be a large payload on the upper row. Because PASCAL's mission objectives involve sample collection we are sensitive to contamination from other parts of the HASP payload and to the exhaust of any instruments that sample air. We believe the cleanest sampling during float would occur on the highest row of large payloads. PASCAL will be sampling with POPS on ascent and descent but sample collections at float are the highest priority.

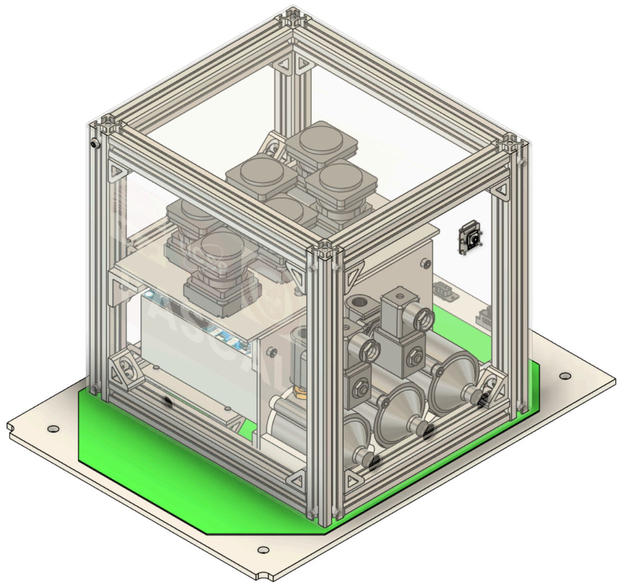
If large payload slots are unavailable, the PASCAL payload could be reduced to the small payloads 3kg and 0.5 Amp constraints and still achieve some of our objectives. While population level analysis, statistical significance of findings, and size differentiation would not be possible, we would still be able to demonstrate a small amount of particle sample collection in the mid-stratosphere and demonstrate performance of small scroll pumps at high-altitudes.

3.8 Special Requests

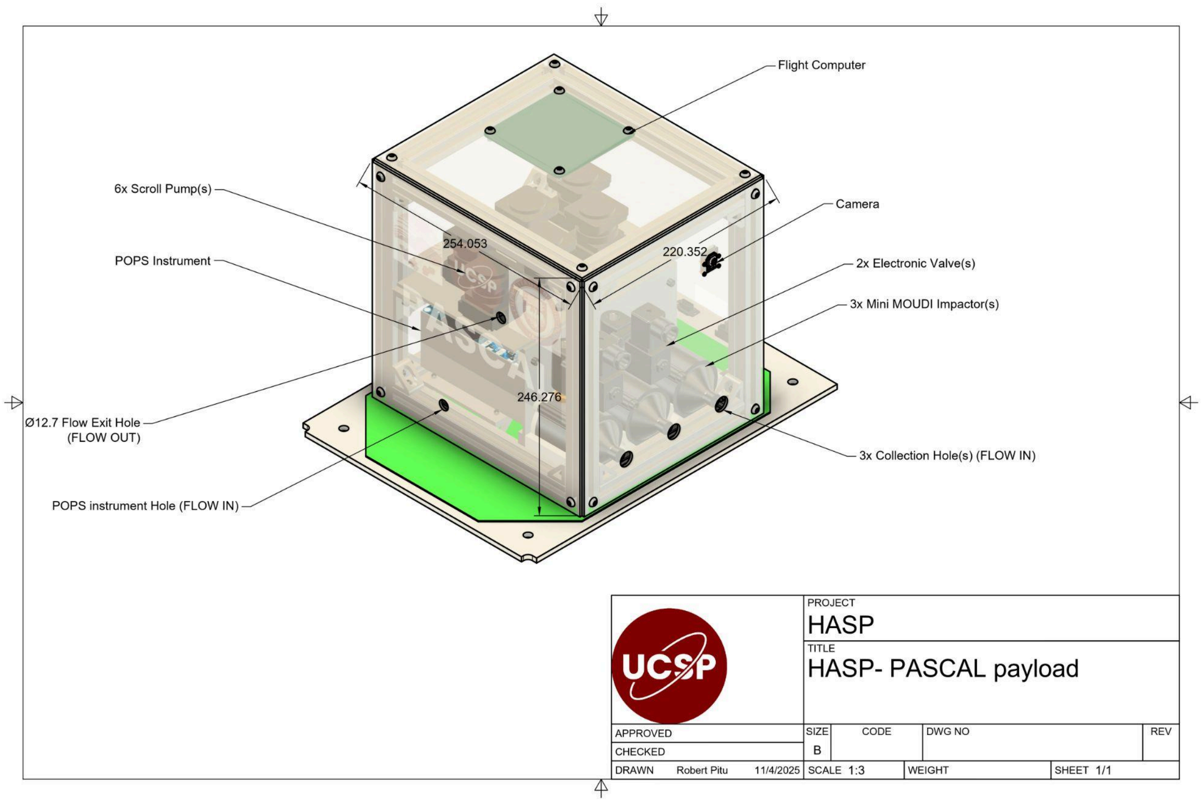
N/A

4 Preliminary Drawings and Diagrams

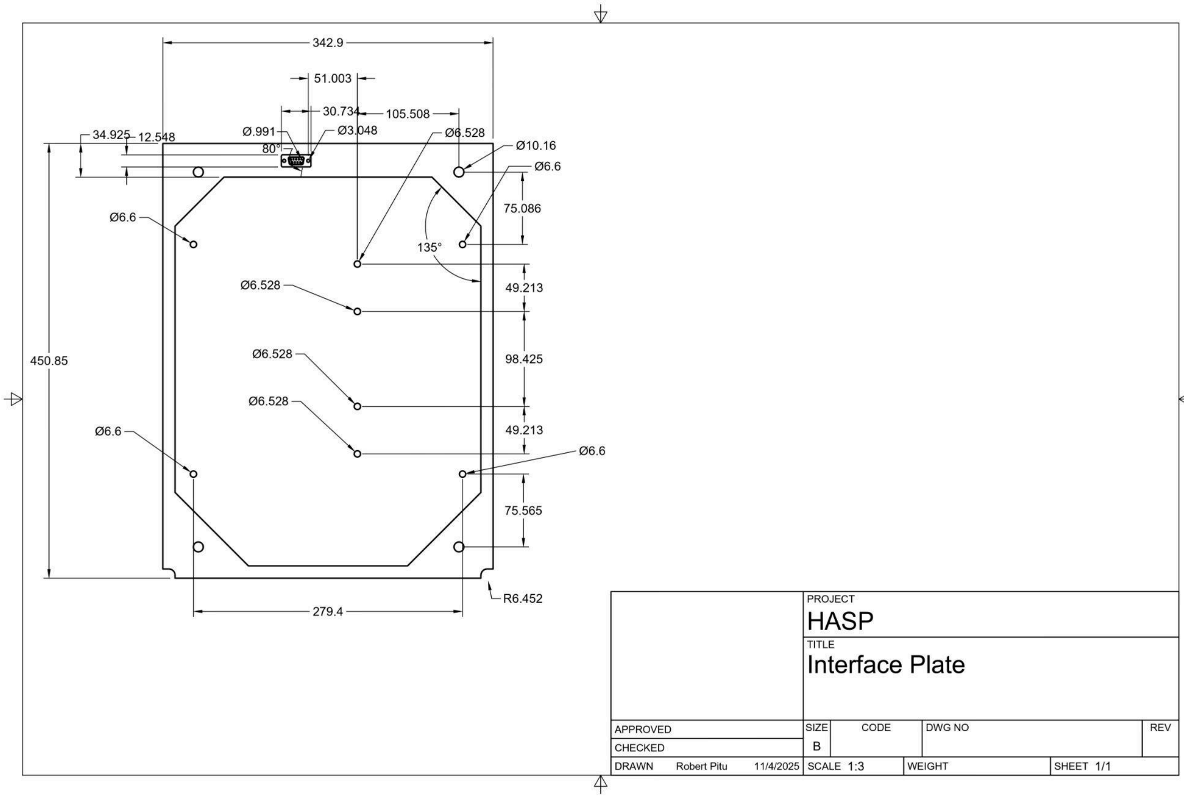
1. Full payload outer enclosure ¾ view



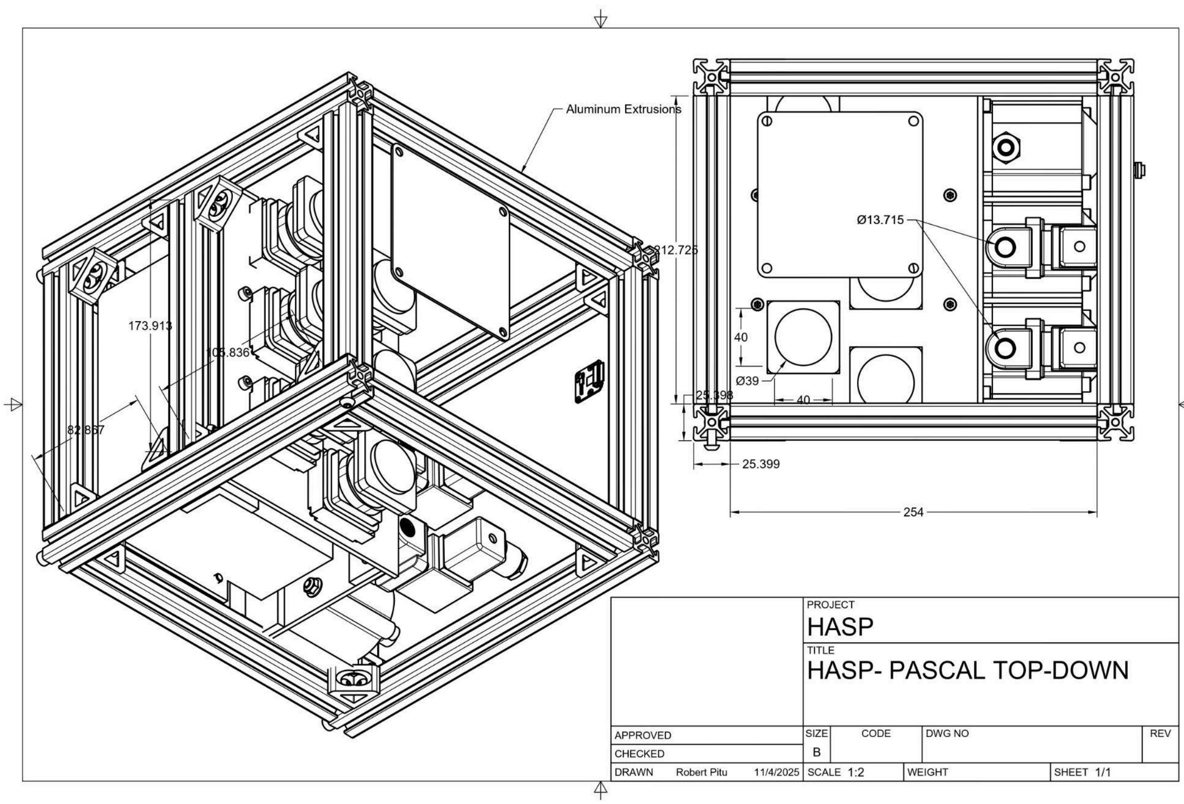
2. Low Opacity for 2-walls of enclosure in ¾ view, showing instrument placement within



3. Zoomed detailing on railings/fixtures to HASP mount/frame



4. Top-Down technical drawing (from fusion of internal placement)



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Appendix A

Dr. Elisabeth Moyer has committed to managing all finances and lab equipment needed for successful completion of this project.

A.1 Avionics Card Detailed Schematics

Due to image size and resolution constraints, we present the schematic broken into several figures.

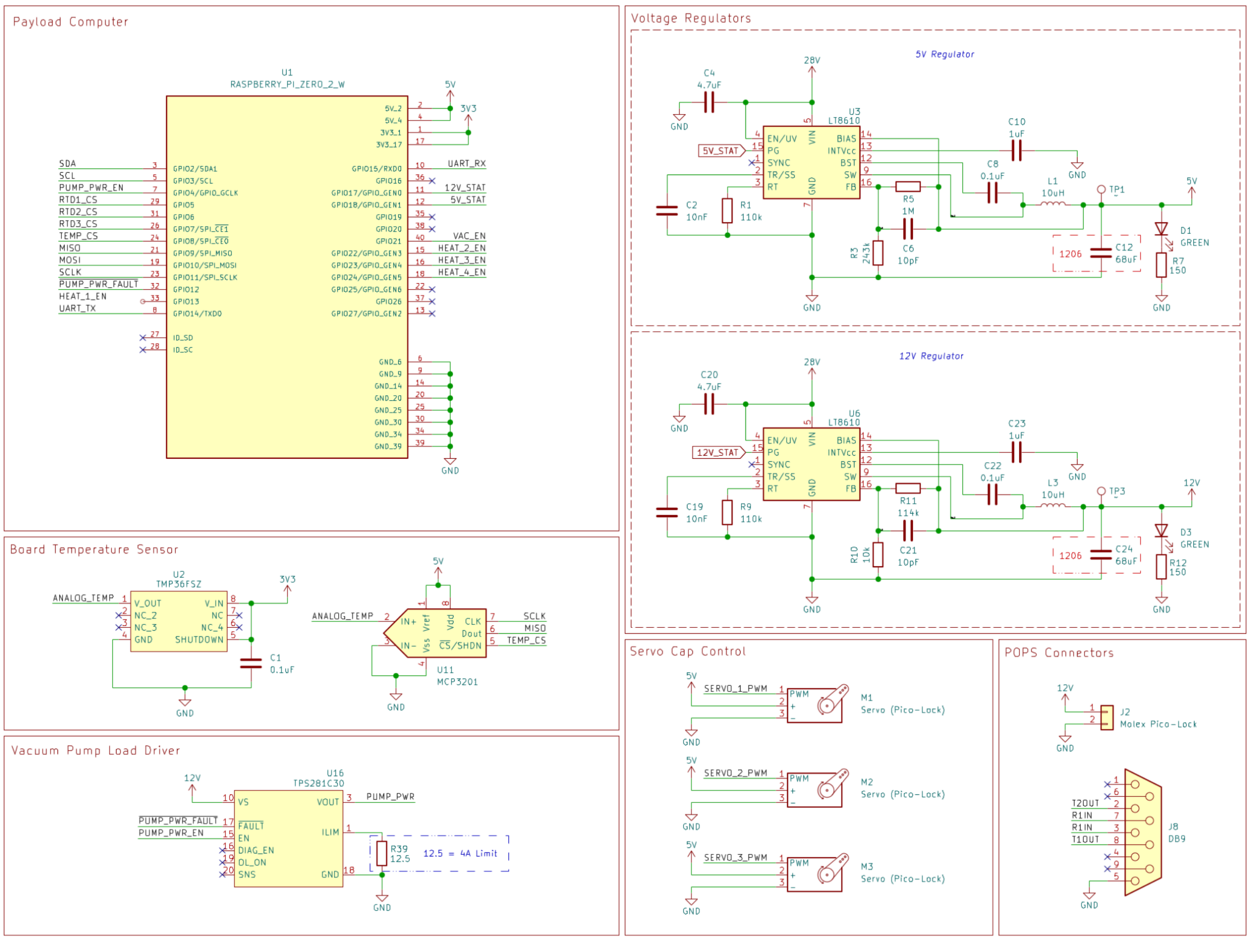

Figure 19: Schematic screenshot with computer, regulators, temp sensor, servo control, and POPS connectors

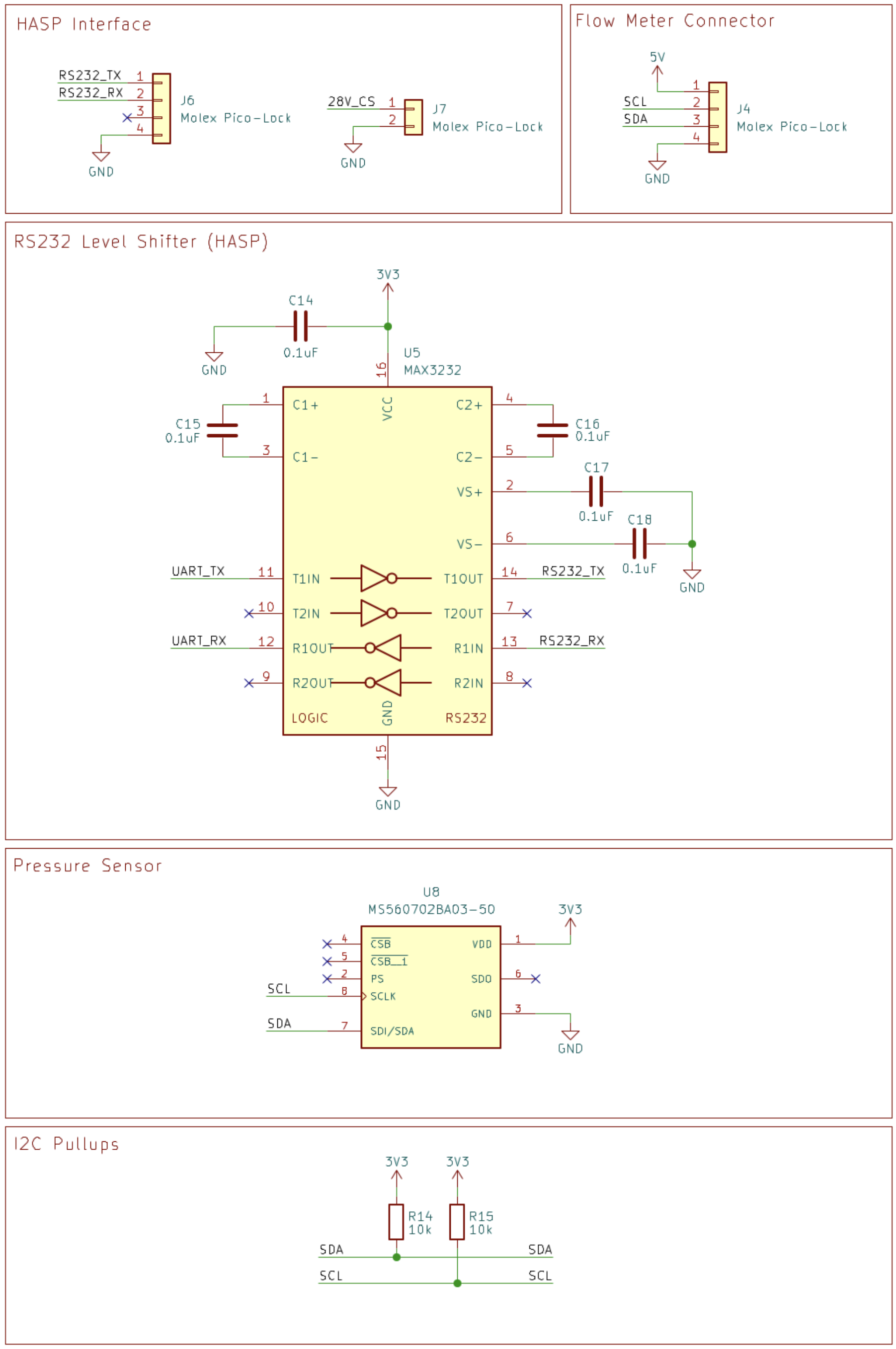

Figure 20: More schematic screenshots including the HASP interface, level shifter, and pressure sensor.

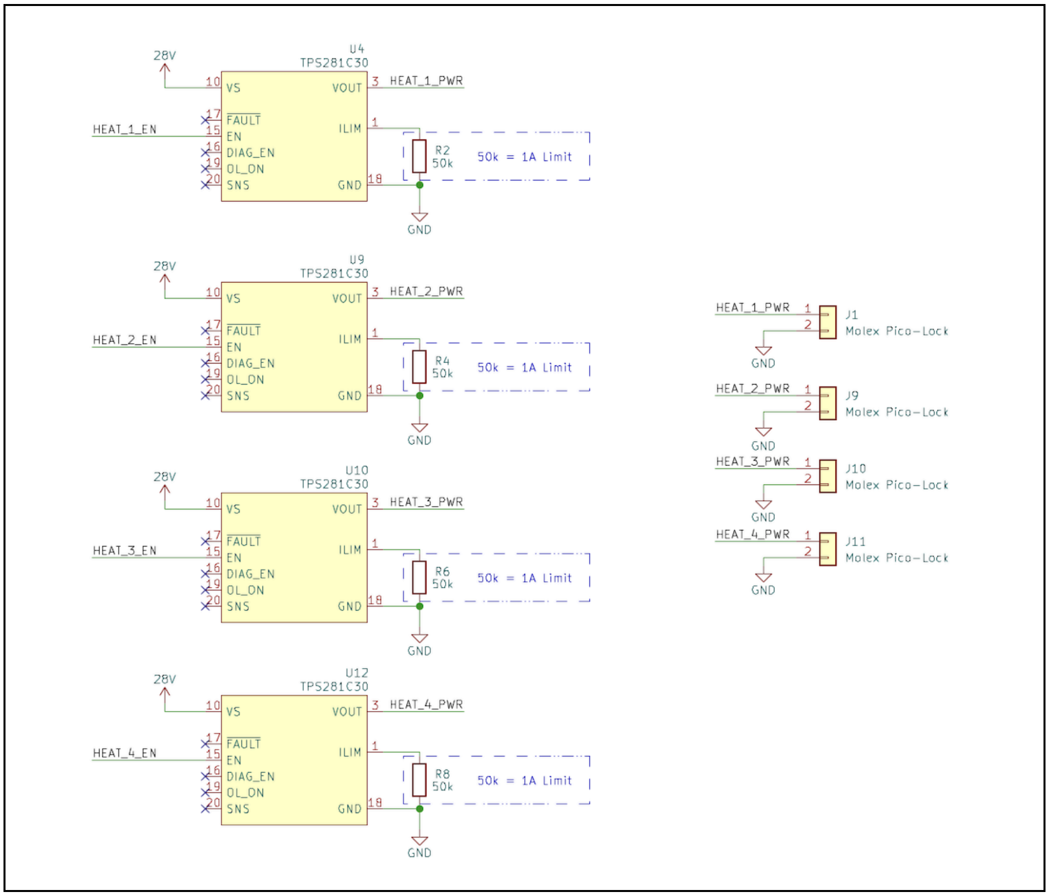

Figure 21: Heater driver array

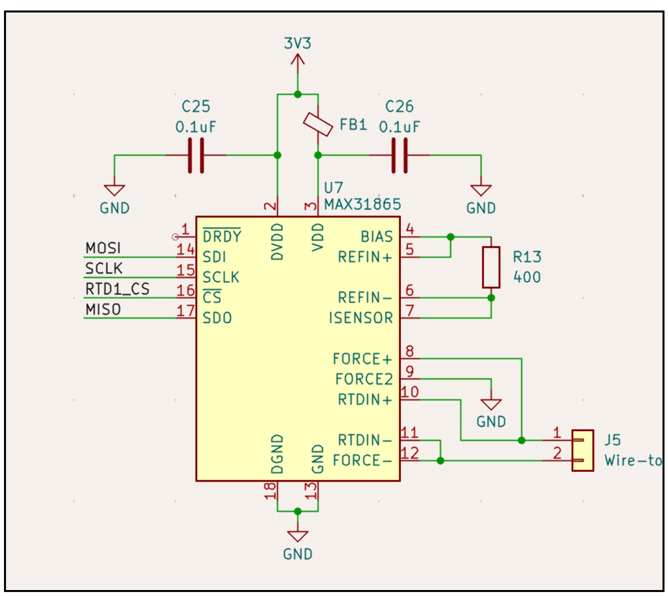

Figure 22: RTD ADC (duplicated 8 times)

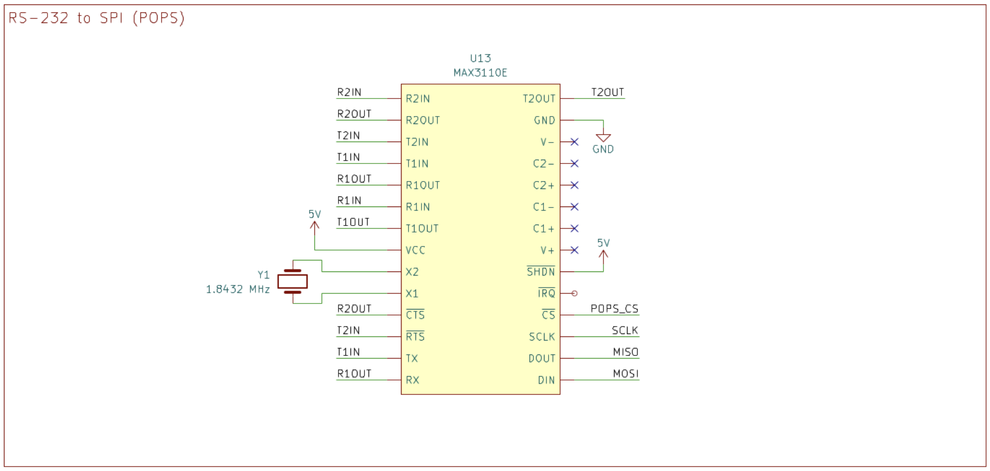

Figure 23: RS-232 to SPI module for communication with POPS

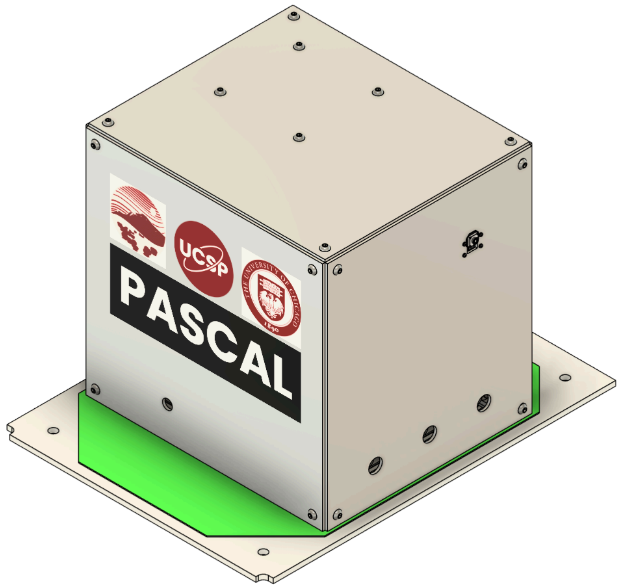

Figure 24: Isometric view of the outside of the PASCAL payload

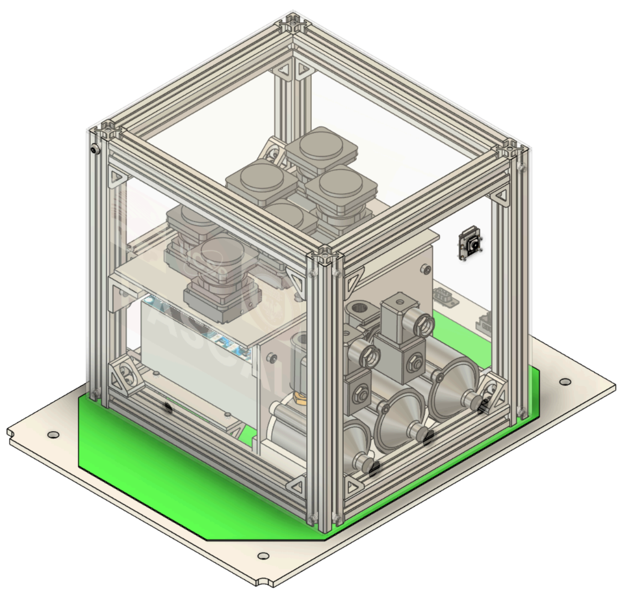

Figure 25: Isometric view of the inside of the PASCAL payload

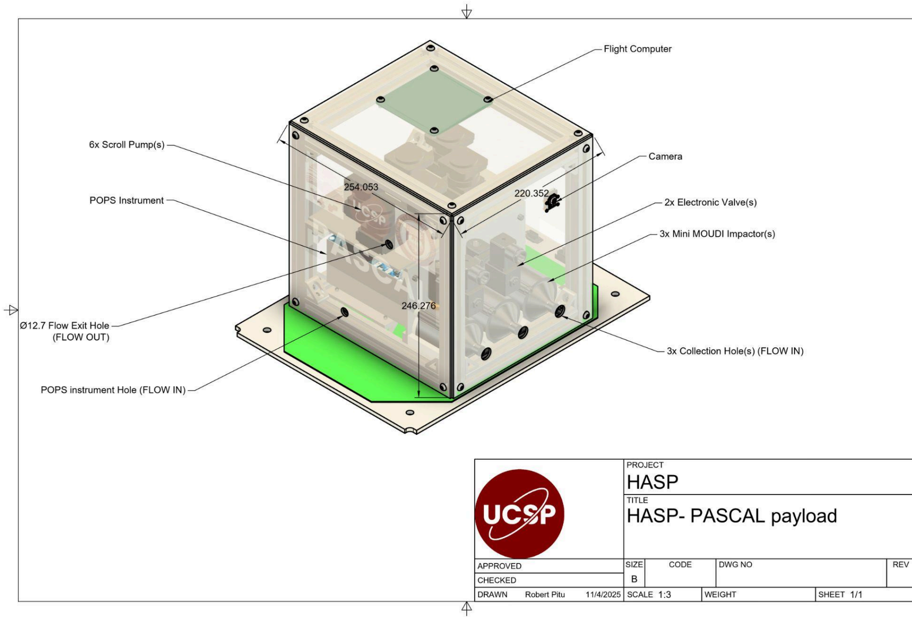

Figure 26: Engineering drawing of PASCAL payload

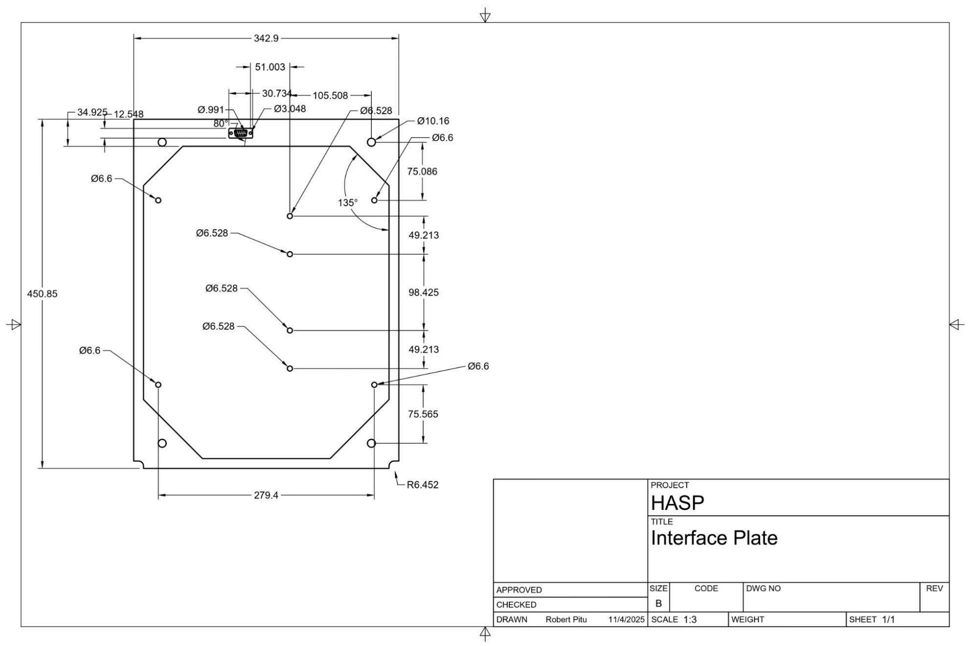

Figure 27. Engineering drawing of the payload interface base plate

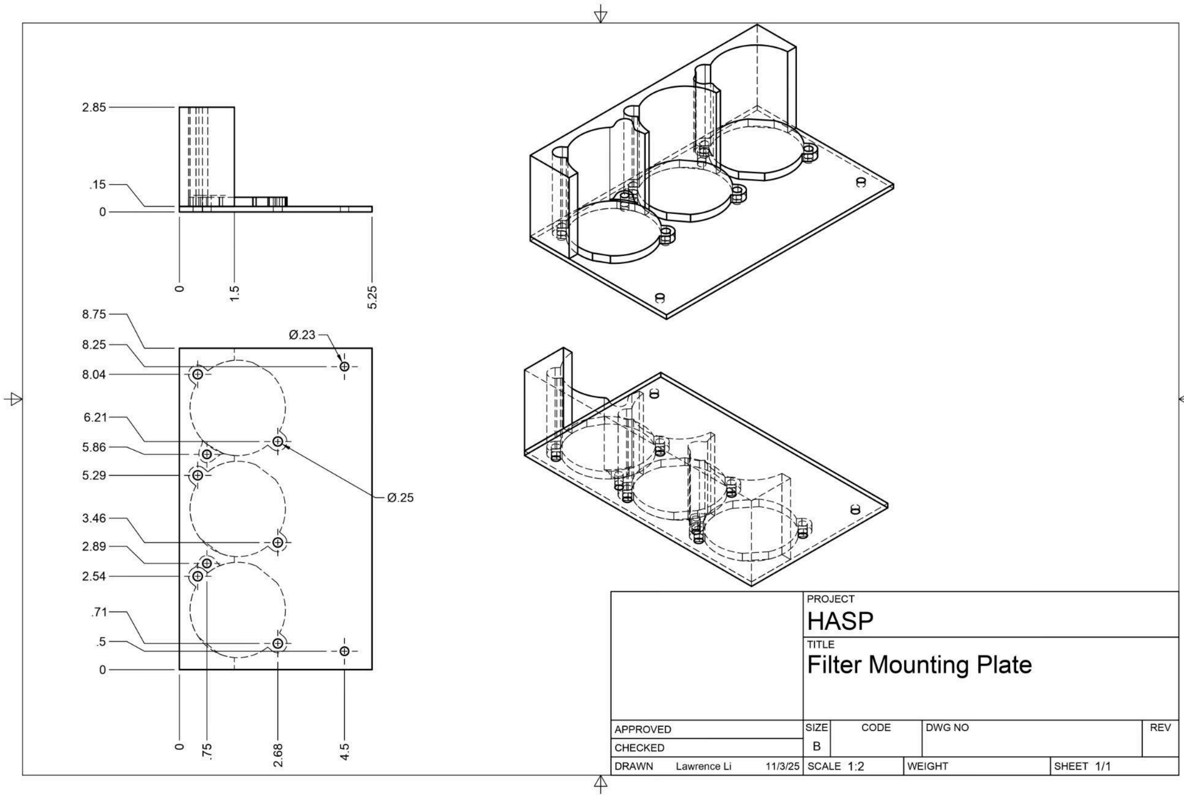

Figure 28. Engineering drawing of the mounting plates for the mini-MOUDI impactors

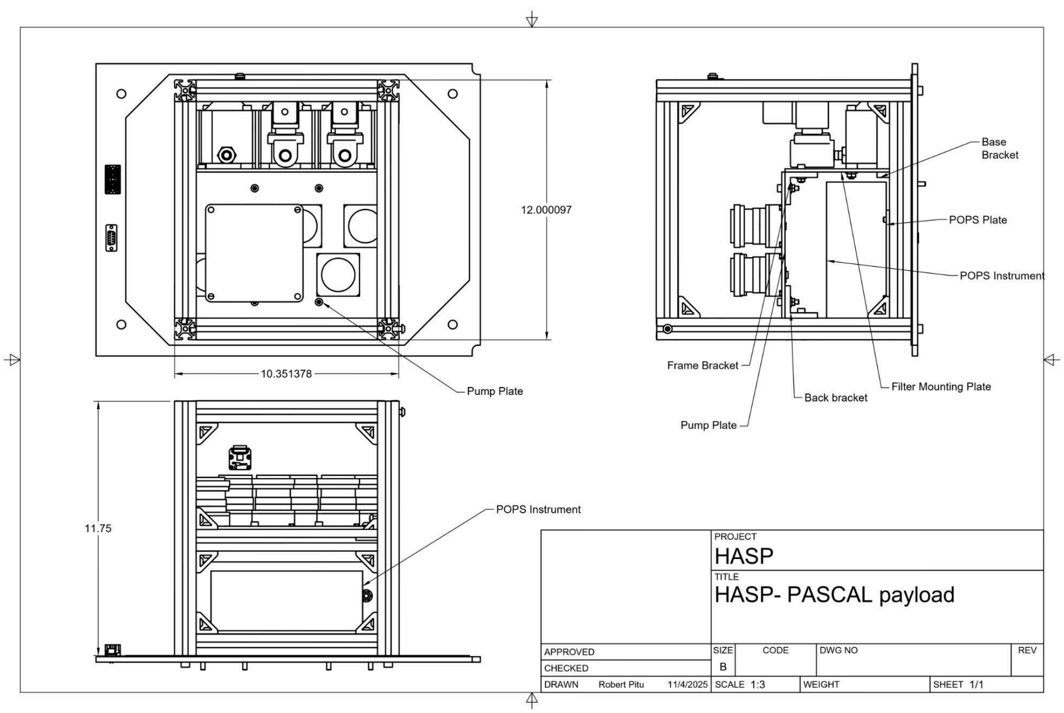

Figure 29. Top and side engineering drawings of the inside PASCAL payload

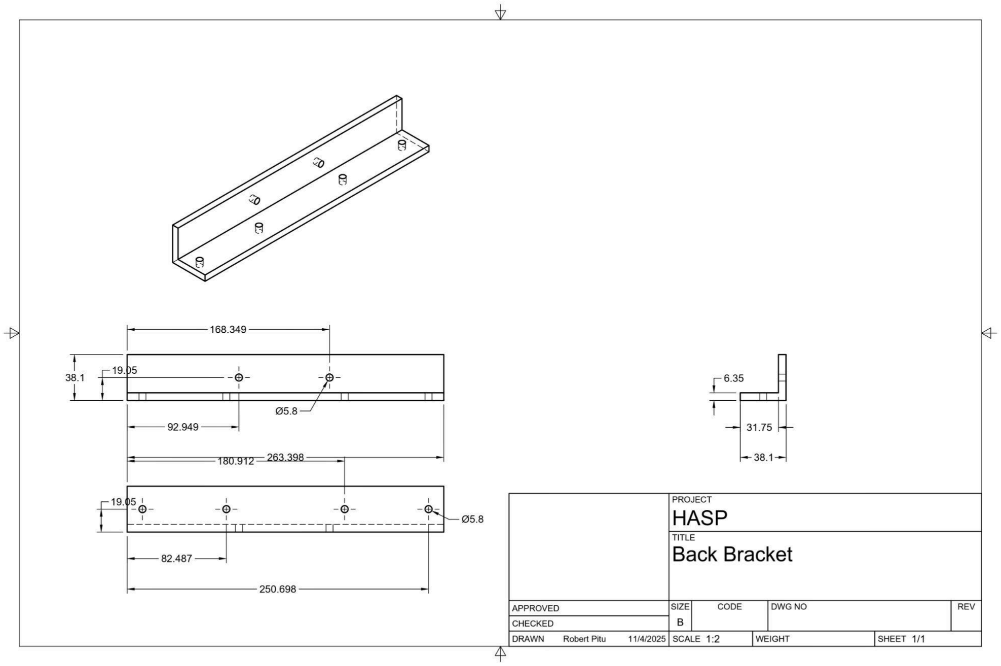

Figure 30. Engineering drawings of the back brackets connecting the vacuum pumps

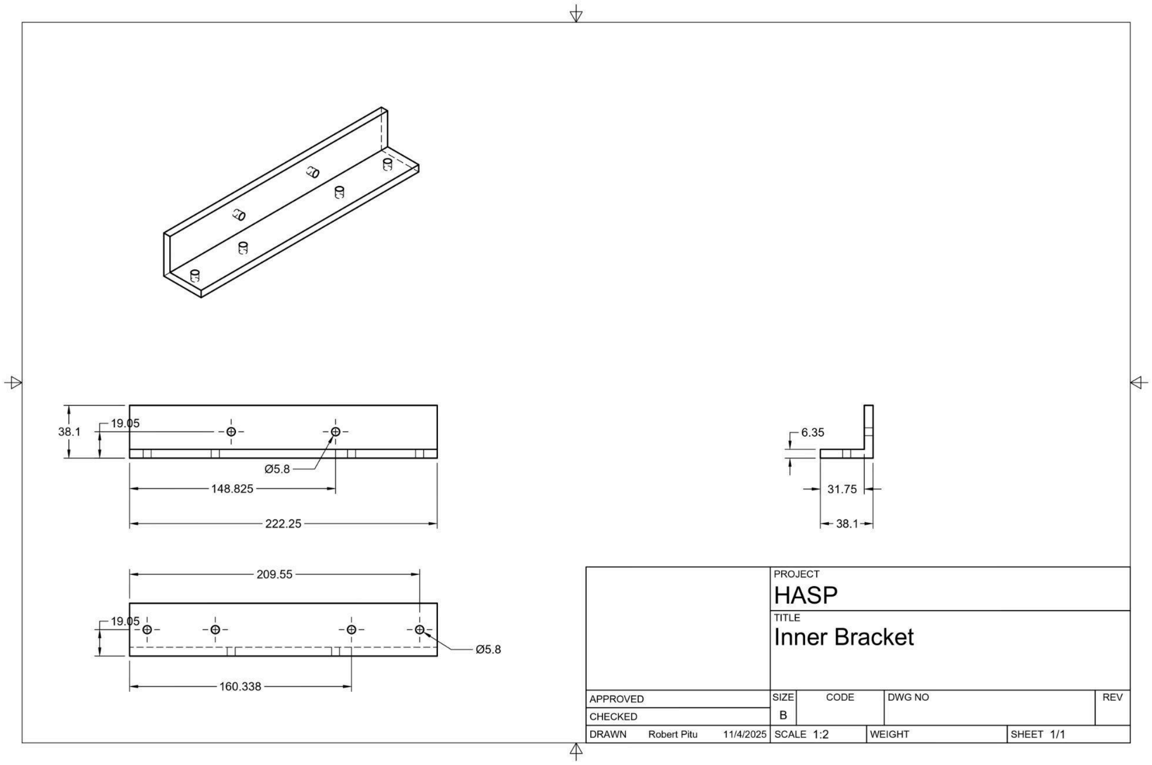

Figure 31. Engineering drawings of the inner payload brackets

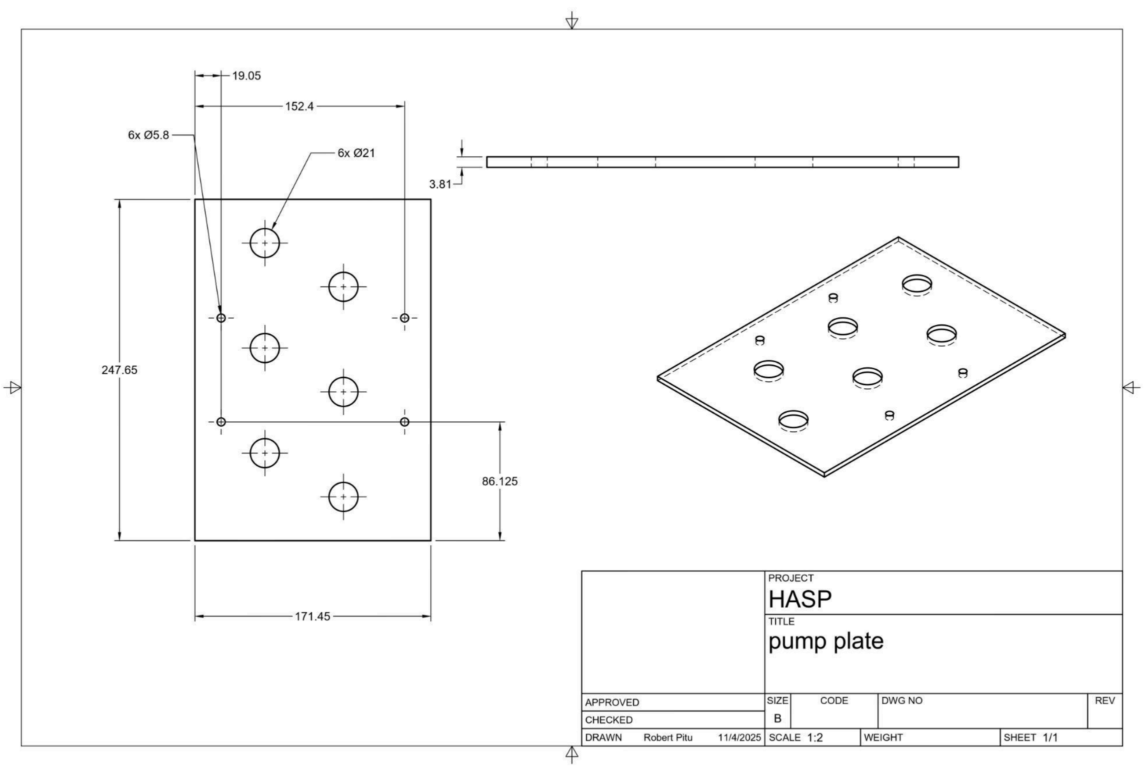

Figure 32. Engineering drawings of the payload mounted vacuum pump plate

Appendix B: NASA Hazard Tables

Appendix B.1 Radio Frequency Transmitter Hazard Docs

This is not applicable to the proposed payload.

HASP 2024 RF System Documentation	
Manufacture Model	
Part Number	
Ground or Flight Transmitter	
Type of Emission	
Transmit Frequency (MHz)	
Receive Frequency (MHz)	
Antenna Type	
Gain (dBi)	
Peak Radiated Power (Watts)	
Average Radiated Power (Watts)	

Appendix B.2 High Voltage Hazard Documentation.

This is not applicable to the proposed payload.

HASP 2024 High Voltage System Documentation	
Manufacture Model	
Part Number	
Location of Voltage Source	
Fully Enclosed (Yes/No)	
Is the High Voltage Source Potted?	

Output Voltage	
Power (W)	
Peak Current (A)	
Run Current (A)	